

LECTURE 1

Sept 30th, 2025

PHASE SPACE: coordinates that describe "dynamic" variables

Dynamical System $\leftrightarrow$ laws of evolution of the phase space

$$\frac{dx}{dt} = \underbrace{v(t, x)}_{\substack{\text{VECTOR} \\ \text{FIELD}}}, \quad x \in \{\text{phase space}\} \subset \mathbb{R}^d.$$

$v(t, x) \in \mathbb{R}^d$
 $\uparrow$
base pt

- Velocity describes the infinitesimal perturbation at x at time t .
- AUTONOMOUS Dynamical System: vector field v is indep. of time
- NON-AUTONOMOUS Dynamical System: vector field varies w/ time.
- Implicitly: $F(x, \dot{x}, \ddot{x}, \dots, \frac{d^k x}{dt^k}) = 0$

$\downarrow$ convert to 1st order

$$\begin{cases} x_0 := x \\ x_1 := \frac{dx}{dt} \\ \vdots \\ x_k := \frac{d^{k-1}x}{dt^{k-1}} \end{cases} \longrightarrow z := \begin{bmatrix} x_0 \\ \vdots \\ x_{k-1} \end{bmatrix} \longrightarrow \frac{dz}{dt} = v(t, z)$$

Note: if $x \in \mathbb{R}^d$, then $z \in \mathbb{R}^{kd}$.

WLOG: we look at 1st order ODEs $\frac{dx}{dt} = v(t, x)$, $x \in \mathbb{R}^d$

Flow $\equiv$ continuous-time dynamical systems.

ODEs

- Initial Value Problem
- single soln $\left[\frac{dx}{dt} = v(t, x), \quad x(0) = a \right]$
- Numerical time integration
- Existence and uniqueness
- Regularity

Dynamics

- All possible solutions to an ODE
- Geometry of phase space?
- Volume of initial conditions
- Ergodic theory: relation of long-time evolution of single solution and volumes in phase space
- Perturbations

SOLUTION OF DYNAMICAL
SYSTEMS

TRAJECTORY (or ORBIT) of
THE DYNAMICS

IDEA: Dynamics deals with the qualitative understanding of the orbit structure.

$$x(t) = \varphi(t, x(0))$$

Semigroup Property: $\varphi(t+s, x(0)) = \varphi(t, \varphi(s, x(0)))$

HOLDS FOR $\nearrow$ i.e., $\varphi^t \circ \varphi^s = \varphi^{t+s} = \varphi^s \circ \varphi^t$.
AUTONOMOUS SYSTEMS

DISCRETE-TIME DYNAMICS: $\frac{dx}{dt} = v(t, x)$ and look at discrete times $x(0), x(1), x(2), \dots$ for $z \in \mathbb{Z}$; i.e., $x(z) = \varphi^z(x(0))$
!!

AUTONOMOUS DYNAMICS: $x_{t+1} = F(x_t)$,

where $F: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is called a self-map

$$x_t = F^t(x_0) = \underbrace{(F \circ \dots \circ F)}_{t \text{ times}}(x_0)$$

$$x_{t+1} = \overset{\substack{\uparrow \\ \text{MAP}}}{F_t} x_0, t \in \mathbb{Z}$$

- EXAMPLES :
- Low-dimensional dynamics ($\dim = 1, 2, 3, 4$)
 - Climate dynamics: - very high-dimensional system ($x \in \mathbb{R}^d, d = O(10^4)$)
 - Chaos: sensitivity to initial conditions.
 - Lorenz Equations $\rightarrow$ came from modeling atmospheric convection.

LINEAR DYNAMICS $\frac{dx}{dt} = Ax$, $x \in \mathbb{R}^d$, $A \in \mathbb{R}^{d \times d}$.

↑
linear operator

Solution: $x(t) = e^{tA} x(0)$.

- CASE 1: A is diagonalizable; i.e., $A = P \Lambda P^{-1}$
- columns are eigenvectors of A
- ↑
 $\text{diag}(\lambda_1(A), \dots, \lambda_n(A))$

LECTURE 2

LINEAR HOMOGENEOUS ODES

Oct 2nd, 2025

LINEAR HOMOGENEOUS ODES

$$(*) \begin{cases} \dot{x} = A(t)x, & x \in \mathbb{R}^n \\ A(t) \in \mathbb{R}^{n \times n} \text{ continuous in time.} \end{cases}$$

$x \equiv 0$ solves $(*)$.

SUPERPOSITION PRINCIPLE: If $x_1(t)$ and $x_2(t)$ are solutions to $(*)$, then $x(t) := c_1 x_1(t) + c_2 x_2(t)$ is also a solution $\forall c_1, c_2 \in \mathbb{R}$.

NOTE: If we can find n lin. indep. solutions $x_1(t), \dots, x_n(t)$ then the general solution to $(*)$ is

$$x(t) = c_1 x_1(t) + \dots + c_n x_n(t).$$

AUTONOMOUS PROBLEM:
$$\begin{cases} \dot{x} = Ax, & x \in \mathbb{R}^n \text{ and } A \in \mathbb{R}^{n \times n} \\ x(0) = x_0 \in \mathbb{R}^n \end{cases}$$

• APPROACH 1: If A has n distinct eigenvalues $\lambda_1, \dots, \lambda_n$ and n lin. indep. eigenvectors $v_1, \dots, v_n$ satisfy the characteristic eq. of A
 $0 \doteq \det(A - \lambda I)$

General Solution:

$$x(t) = \sum_{j=1}^n c_j e^{\lambda_j t} v_j = \begin{bmatrix} e^{\lambda_1 t} v_1 & \dots & e^{\lambda_n t} v_n \end{bmatrix} \underbrace{\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}}_{\text{determined by initial condition}}$$

$$x(0) = x_0 = \begin{bmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{bmatrix}^{-1} x_0$$

↖ invertible b/c eigenvectors v_j are lin. indep

NOTATION: $\Phi(t) := \begin{bmatrix} | & & | \\ e^{\lambda_1 t} v_1 & \dots & e^{\lambda_n t} v_n \\ | & & | \end{bmatrix}$ (FUNDAMENTAL MATRIX SOLUTION)

This solves $\frac{d}{dt} \Phi(t) = A \Phi(t)$. CONVENTION: $\Phi(0) = \mathbb{1}$

UPSHOT: $\dot{x} = Ax$, $x(0) = x_0$ is solved by $x = \Phi(t) x_0$.

ASIDE: • If some λ_j is complex, $\lambda_j = \alpha + i\beta$, then v_j is also complex. Note that $\bar{\lambda}_j = \alpha - i\beta$ is also an eigenvalue with eigenvector $\bar{v}$.

In this case, the real solution involving λ is

$$x(t) = c e^{\lambda t} v_j + \bar{c} e^{\bar{\lambda} t} \bar{v}_j, \quad c \in \mathbb{C}.$$

- If A is defective; i.e., doesn't have n lin. indep. eigenvectors (repeated eigenvalues). Look at generalized eigenvectors.

Ex: λ is an eigenvalue w/ multiplicity 2 and its eigenvector v spans a 1-dim subspace $\leftarrow (A - \lambda \mathbb{1})v = 0$.

Generalized eigenvector $w \leftarrow \begin{cases} (A - \lambda \mathbb{1})^2 w = 0 \\ \& (A - \lambda \mathbb{1})w = v \end{cases}$

One possibility is $(A - \lambda \mathbb{1})w = v \iff Aw = \lambda w + v$.

Solution: $x_1(t) = e^{\lambda t} v$ and $x_2(t) = e^{\lambda t} (tv + w)$
are lin. indep. solutions.

- APPROACH 2: Solution to $\begin{cases} \dot{x} = Ax \\ x(0) = x_0 \end{cases}$ is $x(t) = e^{tA} x_0$.

Recall: $e^{tA} = \sum_{k=0}^{\infty} \frac{(tA)^k}{k!}$.

Check: $x(0) = e^{0A} x_0 = x_0$

$$\begin{aligned}\dot{x} &= \lim_{h \rightarrow 0} \frac{x(t+h) - x(t)}{h} = \lim_{h \rightarrow 0} \frac{e^{(t+h)A} x_0 - e^{tA} x_0}{h} = \lim_{h \rightarrow 0} \frac{e^{hA} - \text{Id}}{h} e^{tA} x_0 \\ &= \left(\lim_{h \rightarrow 0} \frac{e^{hA} - \text{Id}}{h} \right) x(t) = Ax.\end{aligned}$$

since autonomous, they commute. False if non-autonomous.

- CASE 1: Diagonalizable $A \rightsquigarrow A = P \Delta P^{-1}$, where $\Delta = \text{diag}(\lambda_1, \dots, \lambda_n)$, $P = \begin{bmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{bmatrix}$

$$\begin{aligned}\text{Then } x(t) &= e^{tA} x_0 = P \underbrace{e^{t\Delta}}_{e^{t\Delta} = \text{diag}(e^{t\lambda_1}, \dots, e^{t\lambda_n})} P^{-1} x_0\end{aligned}$$

- CASE 2: Defective A . Recall the ^{Jordan} V decomposition of a complex matrix $A = S + N$ where $[S, N] = 0$.

$S = P D P^{-1}$ semisimple (diagonalizable)

N nilpotent

S and N commute

$$\text{Solution: } x(t) = e^{tA} x_0 = e^{t(S+N)} x_0 = e^{tS} e^{tN} x_0$$

$$= \underbrace{P e^{tD} P^{-1}}_{S \text{ diagonalizable}} \left(\sum_{k=0}^{n-1} \frac{(tN)^k}{k!} \right) x_0.$$

N nilpotent w/ nilpotency n

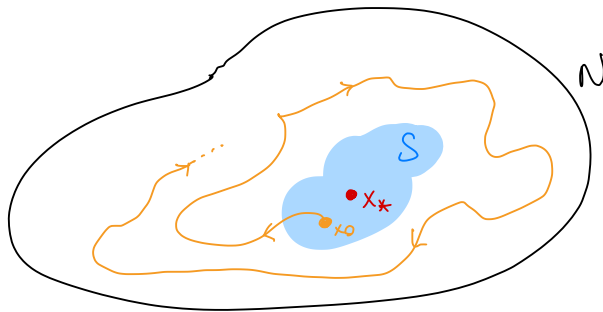
LECTURE 3

Oct 7th, 2025

Motivation: QZ of Thwot asks about the stability of solution $x=0$
to $\dot{x} = A(t)x$.

DEF: We say a pt. x_* is an equilibrium for the system
 $\dot{x} = V(x)$ iff $V(x_*) = 0$.

DEF: We say an equilibrium pt. x_* (e.g., $x_* = 0$) is Lyapunov stable iff for every neighborhood $N \ni x_*$, there exists a neighborhood $S \subset N$ such that, for all $x(0) \in S$, we have $x(t) \in N \ \forall t \geq 0$ (i.e., we never escape).



- x_* is UNSTABLE $\Leftrightarrow x_*$ is not Lyapunov stable.
- x_* is ASYMPTOTICALLY STABLE $\Leftrightarrow x_*$ is Lyapunov stable and
 $\lim_{t \rightarrow \infty} x(t) = x_*$ for $x(t) \in N$

REMARK: For $\dot{x} = Ax$, $x=0$ is asymptotically stable if all eigenvalues $\{\lambda_j\}$ of A have $\text{Re}(\lambda) < 0$.
If $\text{Re}(\lambda) > 0$ for any eigenvalue, then $x=0$ is unstable.

FLOQUET THEORY & MONODROMY MATRIX

Consider $\begin{cases} \dot{x} = A(t)x \text{ s.t. } A(t+T) = A(t) \quad \forall t \geq 0. \\ x(0) = x_0 \end{cases}$

The fundamental matrix solution $\Phi(t)$ is s.t.

$$\frac{d\Phi}{dt} = A(t)\Phi \quad \text{and} \quad \Phi(0) = I. \quad (\Phi \approx \text{propagator})$$

Then, we have $x(t) = \Phi(t)x(0)$.

DEF: The monodromy matrix is defined as $M := \Phi(T)$

⚠ Very useful to know
• the eigenvals. of M

Say M has eigenvalues $\{\mu_j\}$ called FLOQUET MULTIPLIERS.

REMARK: $x_1 = \Phi(T)x_0 = Mx_0$

$\vdots$

$$x(nT) = M^n x_0$$

↖ Strobe light
analogy

$\& |\mu_j| > 1$, then
 $|\mu_j|^n \xrightarrow{n \rightarrow \infty} \infty$
but if $|\mu_j| < 1$
then $|\mu_j|^n \xrightarrow{n \rightarrow \infty} 0$

* If $|\mu_j| < 1 \quad \forall j$, then $x=0$ is asymptotically stable
* If any μ_j has $|\mu_j| > 1$, then $x=0$ is unstable

EXISTENCE & UNIQUENESS FOR ODEs (Ch 3)

Consider the (IVP) $\begin{cases} \dot{x} = V(x) & , \quad V: \mathbb{R}^n \rightarrow \mathbb{R}^n \\ x(0) = x_0 & , \quad x_0 \in \mathbb{R}^n \end{cases}$.

no uniqueness...
THM: (Local Picard) If $V \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ then there is an interval of time containing zero $(-\varepsilon, \varepsilon)$ such that a solution to (IVP) exists and is given by

$$\dot{x}(t) = x_0 + \int_0^t V(x(s)) ds, \quad t \in (-\varepsilon, \varepsilon).$$

! WARNING: This doesn't imply global existence of solution!
Solution can diverge in finite time

E.g.: $\begin{cases} \dot{x} = x^2 \\ x(0) = 1 \end{cases}$ $\frac{dx}{dt} = x^2 \rightarrow \frac{dx}{x^2} = dt \rightarrow \int_{x(0)}^{x(t)} \frac{dx}{x^2} = \int_0^t dt'$
 $\rightarrow \left[-\frac{1}{x}\right]_1^{x(t)} = t \rightarrow x(t) = \frac{1}{1-t}$, which
as $t \rightarrow 1$, gives $x \rightarrow +\infty$.

$\Rightarrow$ Solution only exists for $t < 1$.

THM: Let $a, b > 0$. If $V: \mathcal{B}_b(x_0) \rightarrow \mathbb{R}^n$ is K -Lipschitz,
then (IVP) has a unique solution $x(t)$ for

$$t \in J := [-a, a] := \left[-\frac{b}{M}, \frac{b}{M}\right], \quad M := \max_{x \in \mathcal{B}_b(x_0)} |V(x)|$$

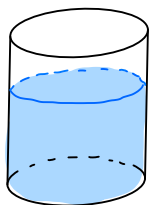
Ensures $x(t)$ doesn't
escape $\mathcal{B}_b(x_0)$ in time $t \in J$

"max speed"

RECALL: $V: B_b(x_0) \rightarrow \mathbb{R}^n$ is K -Lipschitz ∇

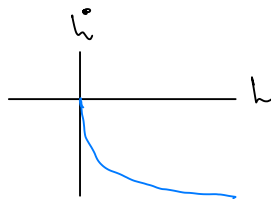
$$\|V(x) - V(y)\| \leq K \|x - y\| \quad \forall x, y \in B_b(x_0).$$

EXAMPLE: (Non-uniqueness in physics)



Punch hole in container at $t=0$. From fluid mechanics

$$\dot{h} = \begin{cases} -\sqrt{h}, & h > 0 \\ 0, & h = 0 \end{cases}$$



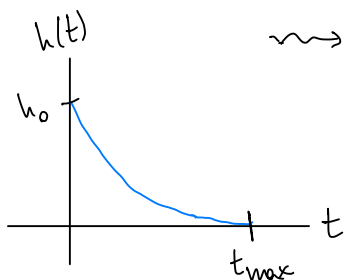
⚠ Not Lipschitz
• at zero

Now, solve for $h_0 > 0$ mindlessly:

$$\frac{dh}{dt} = -\sqrt{h} \quad \rightsquigarrow \quad -\frac{dh}{\sqrt{h}} = dt \quad \rightsquigarrow \quad -\int_{h_0}^{h(t)} \frac{dh}{\sqrt{h}} = \int_0^t ds$$

$$\rightsquigarrow -2\sqrt{h(t)} + 2\sqrt{h_0} = t, \quad t < 2\sqrt{h_0} =: t_{\max}$$

$$\rightsquigarrow h(t) = \begin{cases} (\sqrt{h_0} - t/2)^2, & t < 2\sqrt{h_0} \\ 0, & t \geq 2\sqrt{h_0} \end{cases}$$



how long it takes
to fully drain

⚠ But if you choose your initial condition that gives $h_0 = 0$ (where it's not Lipschitz), there are ∞ -many solutions.

LECTURE 4

Oct 9th, 2025

! **STUDY FOR QUALS.** (proof of existence & uniqueness w/ fixed pts & iterations) and Grönwall's inequality

* TOWARDS THE PROOF OF PICARD'S THM

Rewrite (IVP) as $x(t) = x_0 + \int_0^t v(x(s)) ds$ } Equivalent to (IVP)
and the Contraction Mapping Thm.

DEF: (PICARD'S MAP) $T(u(t)) := x_0 + \int_0^t v(u(s)) ds$

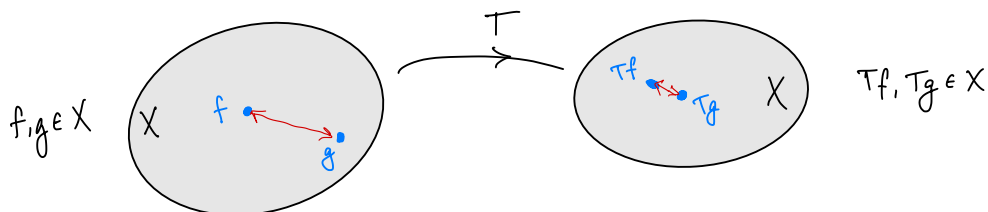
! REMARK: A fixed pt. of T (i.e., x s.t. $Tx = x$) is a solution of (IVP) b/c

$$x(t) = T(x(t)) = x_0 + \int_0^t v(x(s)) ds.$$

PICARD ITERATION: $u_0(t) \xrightarrow{T} u_1(t) \xrightarrow{T} u_2(t) \xrightarrow{T} \dots$
and hope it converges.

CONTRACTION MAPS

Let X be a Banach space. A contraction map T looks like:



← i.e., complete; i.e., Cauchy seqs converge

THM: Let X be a Banach space and let $T: X \rightarrow X$. If T is a contraction (i.e., $\exists c \in (0, 1)$ such that $d = \text{metric on } X \rightarrow d(T(f), T(g)) \leq c d(f, g) \quad \forall f, g \in X$).

Then, T has a unique fixed point $x_* \in X$; i.e., $Tx_* = x_*$.

↪ Time interval in Picard's Thm

PICARD'S MAP: Take $X = C^0(J, B_b(x_0))$ with

$$d(u(t), v(t)) := \sup_{t \in J} |u(t) - v(t)|.$$

Set $T: X \rightarrow X$ given by

$$T(u(t)) = x_0 + \int_0^t V(u(s)) ds.$$

(Note: $T \in \text{Aut}(X)$ b/c $\int \in \text{Aut}(\text{cont. fcts.})$.)

CLAIM 1: $T: X \rightarrow X$ is a contraction.

PF: Let $u_1, u_2 \in X = C^0(J, B_b(x_0))$. Then

$$\begin{aligned} |Tu_1 - Tu_2| &= \left| x_0 + \int_0^t V(u_1(s)) ds - x_0 - \int_0^t V(u_2(s)) ds \right| \\ &= \left| \int_0^t [V(u_1(s)) - V(u_2(s))] ds \right| \\ &\leq \int_0^t |V(u_1(s)) - V(u_2(s))| ds \end{aligned}$$

V is K -Lipschitz

$$\leq \int_0^t K |u_1(s) - u_2(s)| ds$$

Bound the
integral $\rightarrow$

$$\leq K \cdot a \cdot \sup_{t \in J} |u_1(t) - u_2(t)|$$

$$= \underbrace{Ka}_{\text{Take } a \text{ s.t. } 2Ka < 1 \iff a < 1/K} d(u_1, u_2).$$

PF: (PICARD) By Claim 1 above and Banach's fixed pt. thm, we're done.

THM: (Global Existence of Solutions) If $V: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is globally Lipschitz and bounded, then the solutions to

$$(IVP) \begin{cases} \dot{x} = V(x) \\ x(0) = x_0 \end{cases}$$

exist for all $t \in \mathbb{R}$.

Flow Dynamics (ch 4)

exists $\forall t \in \mathbb{R}$

i.e., $t \in \mathbb{R}$

DEF: A complete flow $\varphi_t(x)$ is a 1-parameter mapping $\varphi: \mathbb{R} \times M \rightarrow M$ with the following properties

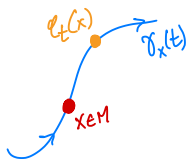
(i) $\varphi_0(x) = x \quad \forall x \in M$; i.e., $\varphi_0 = \text{id}_M$

(Groupoid Property)

(ii) $\varphi_t \circ \varphi_s = \varphi_{t+s} = \varphi_s \circ \varphi_t \Rightarrow \varphi_t$ is invertible: $\varphi_t^{-1} = \varphi_{-t}$.

REMARK: $\varphi_t(x)$ defines a curve $\gamma_x(t)$ in M as t varies;

i.e., $\gamma_x(t) = (t \mapsto \varphi_t(x))$. This $\gamma_x(t)$ is called an ORBIT or TRAJECTORY through x .



EXAMPLE: (1) $\varphi_t(x_0) = x_0 \quad \forall t \in \mathbb{R} \Rightarrow x_0$ is a fixed pt.

(2) If $\exists T > 0$ s.t. $\varphi_T(x_0) = x_0$ and $\varphi_t(x_0) \neq x_0 \quad \forall t \in (0, T)$,

then we have a closed orbit w/ period T

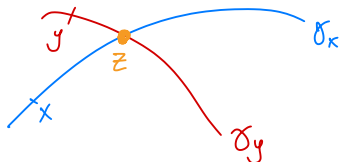


Obs: any pt. in this closed orbit is also T -periodic
b/c $\forall t_* \in (0, T)$

$$\varphi_T(\varphi_{t_*}(x_0)) = \varphi_{t_*}(\varphi_T(x_0)) = \varphi_{t_*}(x_0).$$

(3) The groupoid property implies that two distinct orbits cannot intersect.

Pf: Assume, by contradiction, that they do cross at z . Then



$\varphi_s(y) = z = \varphi_t(x)$ for some t, s . But then

$$\varphi_r(z) = \varphi_{r+t}(x) = \varphi_{r+s}(y) \xrightarrow{\quad // \quad} \text{we assumed } \delta_x \neq \delta_y. \quad \square$$

REMARK: (Connection w/ ODEs) Flows are differentiable mappings (more precisely, local diffeomorphisms) so we can associate a vector field $V \in \mathfrak{X}(M)$ with them via:

$$V(x) = \lim_{t \rightarrow 0} \frac{\varphi_t(x) - x}{t} = \left. \frac{d}{dt} (\varphi_t(x)) \right|_{t=0}$$

Thus, $\varphi_t(x)$ solves the following (IVP)

$$\frac{d}{dt} (\varphi_t(x_0)) = V(\varphi_t(x_0)) \quad , \quad \varphi_0(x_0) = x_0.$$

LECTURE 5

Oct 14th, 2025

STABILITY OF LINEAR SYSTEMS

Recall: $T: (X, d) \rightarrow (X, d)$ is a contraction if $\exists q \in (0, 1)$ s.t.

$$d(T(x), T(y)) \leq q d(x, y) \quad \forall x, y \in X.$$

Note:

$$d(T^k(x), T^k(y)) \leq q^k d(x, y). \quad \text{exponentially fast}$$

Since $0 < q < 1$, as $k \rightarrow \infty$, the orbits converge $\checkmark$ to a fixed pt. of T .
 $\hookrightarrow \{T^k(x)\}_{k \in \mathbb{N}}$ is an orbit of x .

Thm: (Banach Contraction Fixed Pt.) Any contraction in a Banach space has a unique fixed point.

PF: Since (X, d) is Banach (i.e., complete; i.e., every Cauchy sequence converges), it suffices to show that orbits of any $x \in X$ are Cauchy sequences.

Indeed, let $m \geq n$,

$$d(T^m(x), T^n(x)) \leq \sum_{z=n}^{m-1} d(T^{z+1}(x), T^z(x))$$

Triangle
Inequality

$$\stackrel{\substack{T \text{ is a } q\text{-contraction}}}{\leq} \sum_{z=n}^{m-1} q^z d(T(x), x)$$

$$\leq \frac{q^n}{1-q} d(T(x), x)$$

can make this
arbitrarily small
bk $q \in (0, 1)$

$\Rightarrow \{T^k(x)\}_{k \in \mathbb{N}}$ is Cauchy, hence converges to the fixed point x_* of T .

Take $m \rightarrow +\infty$, then

$$d(x_*, T^n(x)) \leq q^n C, \quad C \in \mathbb{R} \text{ constant}$$

$\Uparrow$ The orbit of any x approaches x_* exponentially fast.

REMARK: If $F: X \rightarrow X$ is s.t. $\|DF(x)\| < 1$ and $\sup_{x \in X} \|DF(x)\| < 1$, then, by MVT, F is a contraction

$$\|F_x - F_y\| \leq \|DF(\xi)\| \|x - y\|.$$

REMARK: Take $A \in \mathbb{R}^{n \times n}$, $x \in \mathbb{R}^n$, $x_{t+1} = Ax_t$. Then

$$F_A(x) := Ax$$

is a contraction if $|\lambda_j| < 1 \quad \forall \lambda_j \in \text{Spec}(A)$. Then the fixed pt. is $x = 0$.

REMARK: For $\dot{x} = Ax$, write the eigendecomposition of A (either diagonalize or Jordan form)

$$x(t) = e^{tA} x_0 = P^{-1} e^{t\Lambda} P^{-1} x_0$$

Thus, A is a contraction if all $\lambda_j \in \text{Spec}(A)$ are s.t. $\text{Re}(\lambda_j) < 0$.

———— // ————

LINEAR STABILITY ANALYSIS

Idea: study orbits infinitesimally close to fixed pts.

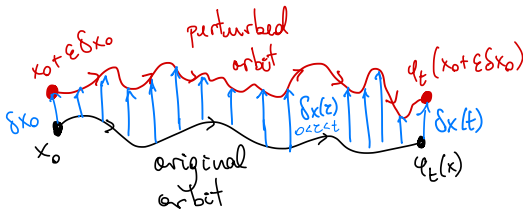
Consider problems of the form

$$\frac{dx}{dt} = V(x) \quad \longleftrightarrow \quad \frac{d\psi_t(x)}{dt} = V(\psi_t(x)), \quad t \in \text{interval of existence}$$

x is initial condition;
i.e., $\varphi_0(x) = x$.

Let δx_0 be a perturbation to initial condition.

Then



$$\delta x(t) = \lim_{\epsilon \rightarrow 0} \frac{\varphi_t(x_0 + \epsilon \delta x_0) - \varphi_t(x_0)}{\epsilon}$$

Now,
$$\frac{d \delta x(t)}{dt} = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(\frac{d \varphi_t}{dt}(x_0 + \epsilon \delta x_0) - \frac{d \varphi_t}{dt}(x_0) \right)$$

(ODE)
$$= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(V(\varphi_t(x_0 + \epsilon \delta x_0)) - V(\varphi_t(x_0)) \right)$$

Assume $V \in C^1$



$$= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(V(\varphi_t(x_0)) + \epsilon \nabla V(\varphi_t(x_0)) \delta x(t) - V(\varphi_t(x_0)) \right)$$

We can Taylor expand:

$$\varphi_t(x_0 + \epsilon \delta x_0) = \varphi_t(x_0) + \epsilon \delta x(t) + O(\epsilon^2)$$

Upshot: FUNDAMENTAL EQUATION OF LINEAR STABILITY ANALYSIS

$$\frac{d \delta x(t)}{dt} = \nabla V(\varphi_t(x_0)) \delta x(t)$$

REMARKS: (1) The orbit $\varphi_t(x_0)$ is fixed

(2) The ODE that describes how the infinitesimal perturbation $\delta x(t)$ changes over time is LINEAR. That is,

$$\frac{dy}{dt} = Y(t, y).$$

(3) The linear stability ODE is not autonomous because the Jacobian $\nabla V(\varphi_t(x_0)) \in \mathbb{R}^{n \times n}$ is changing with time.

(4) If x_0 is a FIXED POINT, then the ODE is autonomous because x_0 is fixed! 

* LINEAR STABILITY ANALYSIS AROUND FIXED POINTS

Fixed point $\iff x_*$ s.t. $x_* = \varphi_t(x_*) \quad \forall t$.

In this case, the perturbation ODE from above is

$$\frac{d\delta x(t)}{dt} = \nabla V(x_*) \delta x(t)$$

Upshot: If $\operatorname{Re}(\lambda) < 0 \quad \forall \lambda \in \operatorname{Spec}(\nabla V(x_*))$, then

$$\delta x(t) \xrightarrow{t \rightarrow \infty} 0 \implies \boxed{x_* \text{ is stable}}$$

since $\delta x(t) = e^{t \nabla V(x_*)} \delta x_0$

⚠ WARNING: This only holds locally around x_* where Taylor expansion is valid.

LECTURE 6

Oct 16th 2025

INFINITESIMAL PERTURBATIONS: Recall $u(t) \in \mathbb{R}^n$ is the infinitesimal perturbation to $\varphi_t(x)$ satisfies

$$\frac{du(t)}{dt} = \nabla V(\varphi_t(x)) u(t) \quad \leftarrow \text{linear}$$

Flow satisfies $\frac{d\varphi_t(x)}{dt} = \nabla V(\varphi_t(x))$, $\varphi_t(x) \in \mathbb{R}^n$.

NOTE: $\{\varphi_t(x)\}_{t \in \mathbb{R}_{>0}}$ is an ORBIT of x .

SPECIALIZED EQUATION TO FIXED POINTS: if $\varphi_t(x_*) = x_* \quad \forall t \geq 0$, we say x_* is a fixed point of V . So, perturbations of fixed points satisfy the following AUTONOMOUS linear system

$$\frac{du(t)}{dt} = \nabla V(\varphi_t(x_*)) u(t) = \nabla V(x_*) u(t)$$

$$\text{i.e., } u(t) = e^{t \nabla V(x_*)} u(0),$$

$\nabla V(x_*) \equiv$ Jacobian of matrix $V: \mathbb{R}^n \rightarrow \mathbb{R}^n$ at
fixed point x_* .

Jacobian

Remark: • if, $\forall \lambda \in \text{Spec}(\nabla V(x_*))$, we have $\text{Re}(\lambda) < 0$,
 then x_* is asymptotically stable since

$$\|u(t)\| \leq C e^{-\alpha t} \|u(0)\|$$

for some $\alpha, C > 0$.

- if at least one $\lambda \in \text{Spec}(\nabla V(x_*))$ have $\text{Re}(\lambda) > 0$, then,
 for almost every $u(0)$,

$$\|u(t)\| \xrightarrow{t \rightarrow \infty} \infty$$

$$u(t) = e^{t \nabla V(x_*)} u(0) = P^{-1} e^{t \Lambda} P u(0)$$

DEF: A fixed point x_* is called hyperbolic fixed point
 if $\text{Re}(\lambda) \neq 0 \quad \forall \lambda \in \text{Spec}(\nabla V(x_*))$.

RECALL: (Aside on maps) $x_{t+1} = F(x_t)$ (map)

$$x_{t+1} = DF(x_t) \text{ (linearization)}$$

$$u_t = (DF(x_*))^t u_0$$

- $\text{Re}(\lambda) > 1$ even for one eigenvalue λ of $DF(x_*)$

$\Downarrow$

almost all solutions u_t

If v_λ is eigenvector of
 eigenval. λ , $\text{Re}(\lambda) < 1$,

then $v_\lambda \rightarrow 0$, but choosing v_λ w/ $P = 0$.

blow up?

REMARK: The behavior of the nonlinear system is described "locally" by its linearization around fixed points.

* ASYMPTOTIC STABILITY: We say that a fixed point x_* ($\varphi_t(x_*) = x_*$) is asymptotically stable iff there exists a nbhd U of x_* s.t. $\varphi_t(x) \xrightarrow{t \rightarrow \infty} x_* \quad \forall x \in U$.

* LINEAR STABILITY: $\operatorname{Re}(\lambda) < 0 \quad \forall \lambda \in \operatorname{Spec}(DV(x_*))$

THM: If $V \in C^1$ and $\exists x_*$ s.t. $V(x_*) = 0$, then
linear stability $\Rightarrow$ Asymptotic stability.

Pf: Given $\frac{d\varphi_t(x)}{dt} = V(\varphi_t(x))$. Say x_* is a fixed pt. of

φ_t i.e., $\varphi_t(x_*) = x_* \quad \forall t$. Let $y(t) := \varphi_t(x) - x_*$.

Note

$$\frac{dy}{dt} = V(\varphi_t(x)) = V(y(t) + x_*)$$

$$\begin{aligned} & \text{Taylor Expansion} \rightarrow = \cancel{V(x_*)} + DV(x_*)y(t) + \underbrace{N(y(t))}_{\substack{\text{Nonlinear} \\ \text{terms in } y.}} \end{aligned}$$

$\xrightarrow{=0} \text{ b/c } x_* \text{ is fixed pt.}$

Assumption, take $N(y(t)) = V(y(t) + x_*) - DV(x_*)y(t)$

$$\Rightarrow \frac{dy}{dt} = DV(x_*)y(t) + N(y(t))$$

If $\forall \epsilon \in \mathbb{C}^1$, then $N(y(t)) = o(y(t))$

$$\frac{N(y(t))}{y(t)} \leq c y(t)$$

!! IMPORTANT !!

DUHAMEL'S FORMULA: The solution to an equation of the form

$$\frac{dy}{dt} = Ay(t) + N(y(t))$$

is

$$y(t) = e^{tA} y(0) + \int_0^t e^{(t-z)A} N(y(z)) dz.$$

By assumption of linear stability, $\operatorname{Re}(\lambda) < 0 \quad \forall \lambda \in \operatorname{Spec}(\underbrace{DV(x_*)}_{A})$:
i.e., $\|e^{tA} y(0)\| \leq C e^{-\alpha t} \|y(0)\|$. So, taking norms in Duhamel's formula,

$$\begin{aligned} \|y(t)\| &= \left\| e^{tA} y(0) + \int_0^t e^{(t-z)A} N(y(z)) dz \right\| \\ &\leq \|e^{tA} y(0)\| + \left\| \int_0^t e^{(t-z)A} N(y(z)) dz \right\| \\ &\leq C e^{-\alpha t} \|y(0)\| + \int_0^t \|e^{(t-z)A} N(y(z))\| dz \\ &\leq C e^{-\alpha t} \|y(0)\| + \int_0^t e^{(t-z)A} \|N(y(z))\| dz \end{aligned}$$

Now, by differentiability of V , $\forall \kappa > 0, \exists \delta > 0$ s.t. $\|V(y(z))\| \leq \kappa \|y(z)\|$ whenever $\|y(z)\| < \delta$.
 and continuity of $t \mapsto y(t)$ means κ and δ don't depend on z

Then

$$\|y(t)\| \leq C e^{-\alpha t} \|y(0)\| + C \kappa \int_0^t e^{-\alpha(t-z)} \|y(z)\| dz \quad (*)$$

! IMPORTANT: MEMORIZE FOR QUAL !

GRÖNWALL'S INEQUALITY

If $\beta \in C^1([0, T])$ and

$$\beta(t) \leq C + \int_0^t \gamma(z) \beta(z) dz, \text{ where } \gamma(z) > 0$$

then

$$\beta(t) \leq C e^{\int_0^t \gamma(z) dz} \quad \forall t \in [0, T].$$

EXERCISE: PROVE THIS !

So, multiplying $e^{\alpha t}$ on both sides of $(*)$, we get

$$\underbrace{e^{\alpha t} \|y(t)\|}_{=\beta(t)} \leq \underbrace{C \|y(0)\|}_{\text{constant}} + \int_0^t \underbrace{C \kappa}_{\gamma(z)} \underbrace{e^{\alpha z} \|y(z)\|}_{\beta(z)} dz.$$

So, by Grönwall's inequality,

$$e^{\alpha t} \|y(t)\| \leq C e^{C \kappa t} \|y(0)\|$$

SIGN
ERROR SNAKE

$$\|y(t)\| \leq C e^{t(Ck - \alpha)} \|y(0)\|$$

$\Rightarrow Ck - \alpha < 0$ if $k < \alpha/C$. Then $\|y(t)\| \xrightarrow{t \rightarrow \infty} 0$.

$$\|\varphi_t(x) - x_*\| \rightarrow 0 \text{ as } t \rightarrow \infty$$

$$\Rightarrow \alpha < -\operatorname{Re}(\lambda) \quad \forall \lambda \in \operatorname{Spec}(A).$$

THM: (HARTMAN-GROSMAN) Let x_* be a hyperbolic equilibrium of a C^1 vector field V with flows $\varphi_t(x)$. Then, $\exists$ nbhd $N \ni x_*$ s.t. φ is topologically conjugate to its linearization on N .

LECTURE 7

OCT 21st 2025

STABILITY:

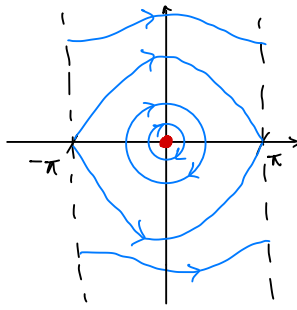
(1) x_* is Lyapunov stable $\Leftrightarrow \forall$ nbhd $N \ni x_*$, $\exists M \subset N$ s.t.
 $\varphi_t(x) \in N \quad \forall t \geq 0, \forall x \in M$.

(2) x_* is asymptotically stable $\Leftrightarrow x_*$ is Lyapunov stable and
 $\varphi_t(x) \xrightarrow{t \rightarrow \infty} x_* \quad \forall x$ close enough to x_* .

EXAMPLES: 1. (IDEAL PENDULUM) $\ddot{\theta} = -\sin \theta$

convert it into a 2×2 system:
$$\begin{cases} \dot{\theta} = y \\ \dot{y} = -\sin \theta \end{cases}$$

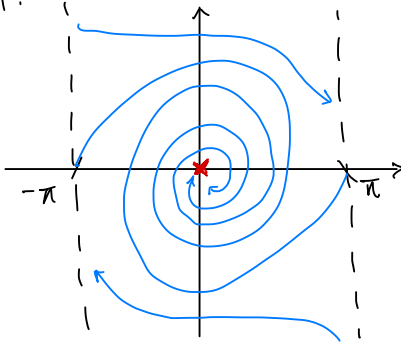
Phase Portrait:



$\theta_x = 0$ is Lyapunov stable
(but not asymptotic)

2. (DAMPED PENDULUM) $\ddot{\theta} = -\sin\theta - \beta\dot{\theta} \Rightarrow \begin{cases} \dot{\theta} = y \\ \dot{y} = -\sin\theta - \beta y \end{cases}$
($\beta > 0$)

Phase Portrait:



$\theta_x = 0$ is both Lyapunov
and asymptotically stable

REMARK: Linearizing is useful when the eigenvalues of the system have non-zero real part.

PROBLEM: Center in the ideal pendulum has both eigenvals. w/ zero real part.

solution ↓

Lyapunov Functions $\simeq$ energy function whose level sets are the orbits on the phase portrait.

Ex: (Undamped Pendulum) $E(\theta, y) = \frac{1}{2}y^2 - \cos\theta$

$$\frac{dE}{dt} = y\dot{y} + \sin\theta y = -y\sin\theta + \sin\theta y = 0$$

Energy is always conserved!

(Damped Pendulum) Same energy $E(\theta, y) = \frac{1}{2}y^2 - \cos\theta$ works.

$$\frac{dE}{dt} = y\dot{y} + \sin\theta y = y(-\sin\theta - \beta y) + \sin\theta y = -\beta y^2 < 0$$

Energy is always lost!

DEF: A continuous function $L: \mathbb{R}^n \rightarrow \mathbb{R}$ is a (strong) Lyapunov function for x_* if $\exists$ open set $U \ni x_*$ where

1) $L(x_*) = 0$

2) $L(x) > 0 \quad \forall x \in U \setminus \{x_*\}$

3) $L(\varphi_t(x)) < L(x) \quad \forall x \in U \setminus \{x_*\}$ and all $t \geq 0$

↑ If " $\leq$ ", we call
 L a weak Lyapunov fct.

THM: Let x_* be an equilibrium of φ_t .

1. If L is a weak Lyapunov function, then x_* is Lyapunov stable.
2. If L is a strong Lyapunov function, then x_* is asymptotically stable.

! CONVERSE is NOT TRUE: Damped pendulum energy $E(\theta, \dot{\theta}) + 1$ is mean Lyapunov even though we know $\theta_* = 0$ is asymptotically stable

Doesn't decay instantaneously along the $y=0$ axis.

Pf: WLOG, $x_* = 0$. Assume L is mean-Lyapunov.

Fix $\varepsilon > 0$ and set $m := \min \{L(x) : |x| \leq \varepsilon\} > 0$.

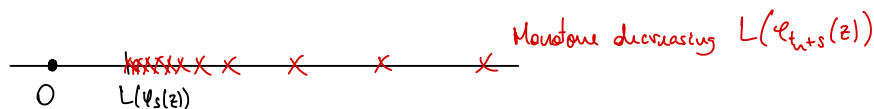
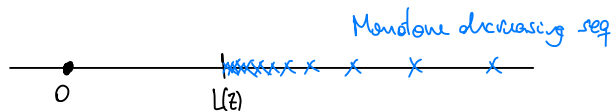
By continuity, $L(x) \xrightarrow{x \rightarrow 0} 0$. So, we can find $\delta > 0$ s.t. $L(x) < m \quad \forall x \in B_\delta(0)$. Since $L(\varphi_t(x)) \leq L(x) < m$ ← Exists b/c the set is closed & bdd
 $\forall x \in B_\delta(0)$, $\varphi_t(x)$ cannot cross $L=m$. Thus, x_* is Lyapunov stable.

Assume L is strong-Lyapunov. By the above, $\forall x \in B_\delta(0)$, $\varphi_t(x) \in B_\varepsilon(0) \quad \forall t \geq 0$.

CLOSED + BOUNDED $\Rightarrow \exists t_n$ s.t. $t_n \rightarrow \infty$, $\varphi_{t_n}(x)$ converges to smth as $n \rightarrow \infty$.
 Bolzano Weierstrass

By contradiction, suppose that $\varphi_{t_n}(x) \xrightarrow{n \rightarrow \infty} z \neq x_* = 0$.

Passing to L , $L(\varphi_t(x))$ is monotone decreasing and has limit $L(z)$ (by continuity).



We get a new seq. $L(\varphi_{s+t_n}(z))$ that is mon. decreasing and has limit $L(\varphi_s(z))$. So, have $t_m > t_n + s$, we get

$$L(z) < L(\varphi_{t_m}(x)) < L(\varphi_{t_n+s}(x)) < L(z)$$

↔ //

← Usually really hard to find...

EASY LYAPUNOV FUNCTIONS:

1. $\dot{x} = -\nabla F(x)$. If x_* is a local minimum of $F(x)$, then it is stable and F is its Lyapunov function.

$$\frac{d}{dt} F(x(t)) = \nabla F(x(t)) \dot{x}(t) = -|\nabla F(x)|^2$$



2. $\ddot{x} = -\nabla F(x)$. Can always rewrite this as

$$\begin{cases} \dot{x} = y \\ \dot{y} = -\nabla F \end{cases}$$

$E(x, y) = \frac{1}{2} |y|^2 + F(x)$ is the Lyapunov function.

3. (HAMILTONIAN SYSTEM) $\begin{cases} \dot{x} = \frac{\partial H}{\partial y} \\ \dot{y} = -\frac{\partial H}{\partial x} \end{cases}$ Hamiltonian is conserved

$$\frac{dH}{dt} = \frac{\partial H}{\partial x} \dot{x} + \frac{\partial H}{\partial y} \dot{y} = 0$$

$\Rightarrow$ Hamiltonian H is Lyapunov fct. near equilibrium of H .

THM: (LASALLE'S INVARIANCE PRINCIPLE) Suppose x_* is an equilibrium for $\dot{x} = f(x)$ and L is non-decreasing on a compact, forward-invariant set $U \ni x_*$.

Let $Z = \{x \in U : \frac{dL}{dt} = 0\}$. Then, if x_* is the largest forward invariant subset of Z , x_* is asymptotically stable.

LEMMA: An asymptotically stable linear system has a strong Lyapunov function of the form

$$L = x^T S x$$

for some symmetric positive-definite S . $\begin{pmatrix} S^T = S \\ S > 0 \end{pmatrix}$

Pf: $\dot{x} = Ax$. We want the decreasing property of L . So,

$$\begin{aligned} \frac{dL}{dt} < 0 &\iff \frac{dL}{dt} = \dot{x}^T S x + x^T S \dot{x} \\ &= x^T A^T S x + x^T S A x \\ &= x^T (A^T S + S A) x \end{aligned}$$

Would be done if $A^T S + S A = -\mathbb{I}$

$$\iff e^{A^T t} (A^T S + S A) e^{A t} = -e^{A^T t} e^{A t}$$

$$\Leftrightarrow \underline{A^T e^{A^T t}} S e^{At} + e^{A^T t} \underline{S A e^{At}} = -e^{A^T t} e^{At}$$

$$\Leftrightarrow \frac{d}{dt} (e^{A^T t} S e^{At}) = -e^{A^T t} e^{At}$$

Integrate

$$\Leftrightarrow e^{A^T t} S e^{At} = C - \int_0^t e^{A^T z} e^{Az} dz$$

Exists b/c we know A is stable $\rightarrow$

$$C = \int_0^\infty e^{A^T t} e^{At} dt = \int_0^\infty e^{A^T t} e^{At} dt - \int_0^t e^{A^T z} e^{Az} dz$$

$$= \int_t^\infty e^{A^T t} e^{At} dt$$

Then $S = \int_0^\infty e^{A^T z} e^{Az} dz$ is symm. positive-def.

$$S^T = \left(\int_0^\infty e^{A^T z} e^{Az} dz \right)^T = \int_0^\infty (e^{A^T z} e^{Az})^T dz = S.$$

Let $x \in \mathbb{R}^n$, $x^T S x = \int_0^\infty \underbrace{x^T e^{A^T z} e^{Az} x}_{\geq 0} dz \geq 0.$



LECTURE 8

Oct 23rd, 2025

LONG-TIME BEHAVIOR OF SOLUTIONS (α - and ω -sets)

* α - AND ω -LIMIT SETS : Suppose $\dot{x} = V(x)$.

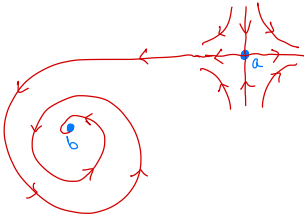
DEF: (ω -limit set)

$$\omega(x) := \{y \in \text{dom } V : \exists t_j \nearrow +\infty \text{ s.t. } \lim_{j \rightarrow \infty} x(t_j) = y\}$$

DEF: (α -limit set)

$$\alpha := \{y \in \text{dom } V : \exists t_j \downarrow -\infty \text{ s.t. } \lim_{j \rightarrow \infty} x(t_j) = y\}$$

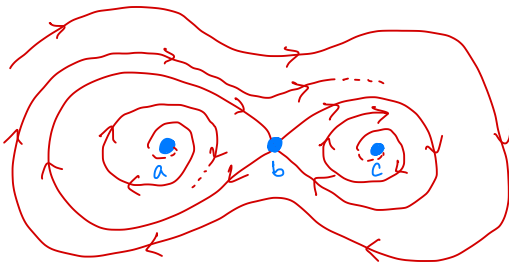
EXAMPLES: 1. Saddle and stable spiral



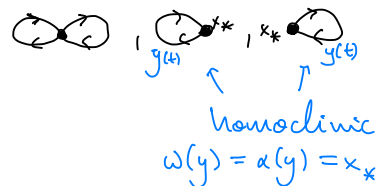
$$\alpha(x) = \{a\}$$

$$\omega(x) = \{b\}$$

2.



The ω -limit sets are



DEF: Solutions connecting different fixed points are called HETEROCLINIC orbits and solutions connecting the same point are called HOMOCLINIC orbits.



PROPERTIES OF LIMIT SETS

1. (Existence) ω -limit set of a bounded orbit is always non-empty. (Heine-Borel)
2. (Closedness) ω -limit sets are closed.
3. (Invariance) If $x \in \omega(x_0)$, then $\varphi_t(x) \in \omega(x_0) \forall t \geq 0$.
4. (Connectedness) ω -limit set of bounded orbit is connected
5. (Transitivity) If $z \in \omega(y)$ and $y \in \omega(x)$, then $z \in \omega(x)$.

THM: (POINCARÉ - BENDIXSON) For planar systems

$\dot{x} = V(x)$, $x \in \mathbb{R}^2$, V is a vector field in $\mathbb{R}^2$, that generate $\varphi_t(x)$, if fixed points are isolated and $\varphi_t(x)$ is bounded. Then, either

1. $\omega(x)$ is an equilibrium, or



2. $\omega(x)$ is a periodic orbit, or



3. for each $u \in \omega(x)$, we have that $\alpha(u)$ and $\omega(u)$ are equilibria.



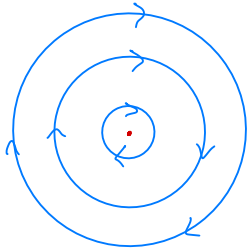
Heteroclinic



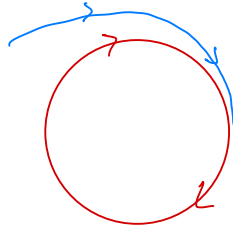
Homoclinic

* LIMIT CYCLES

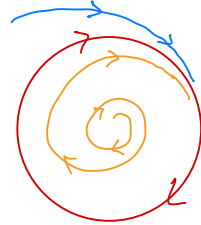
- A periodic orbit γ that is the ω -limit or α -limit set of a point $x \notin \gamma$.



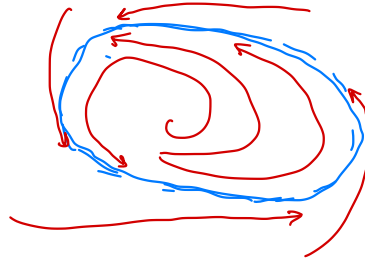
Periodic orbit but
no limit cycles



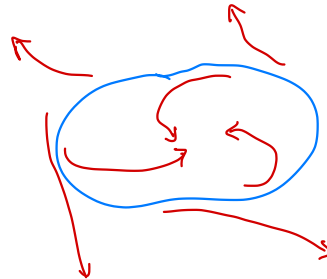
limit cycles
are isolate



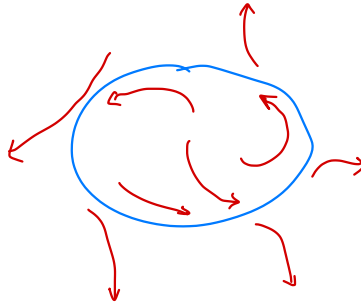
Stable limit cycle $\rightarrow$



Unstable limit cycle $\rightarrow$



Half-stable limit cycle $\rightarrow$



REMARK: Two approaches for linear stability of limit cycle $\gamma(t)$:

LINEARIZE AROUND $\gamma(t)$
(*) ↑
periodic orbit

VS POINCARÉ RETURN MAP
(*)

(*) LINEARIZE AROUND $\gamma(t)$:

$$\begin{cases} \dot{x} = V(x), & x \in \mathbb{R}^n, n \geq 2 \\ V \in C^k(\mathbb{R}^n), & k \geq 1. \end{cases}$$

Let $x(t) := \gamma(t) + \underline{y(t)}$
small perturbation

So, using the ODE,

$V(\gamma(t)) = \dot{\gamma}$ since $\gamma(t)$ solves the ODE

$$\dot{x}(t) = \dot{\gamma}(t) + \dot{y}(t) = V(x(t)) = V(\gamma(t) + y(t))$$

Taylor Expand $\downarrow$

$$= V(\gamma(t)) + \underbrace{D_x V(\gamma(t))}_{\text{Jacobian of } V} y + \text{h.o.t.}$$

linearize $\nearrow$

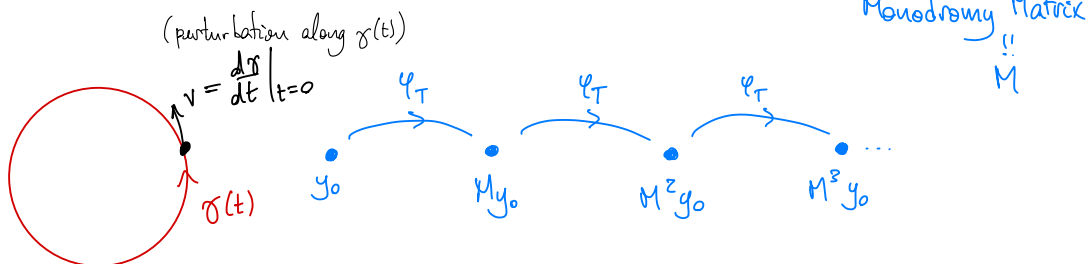
NOTE: $\dot{y} = \underline{D_x V(\gamma(t))} y$

!!
 $A(t) = A(t+T)$

Floquet theory: fundamental matrix solution $\Phi(t)$ is

$$\begin{cases} \frac{d\Phi}{dt} = A(t) \Phi(t), & \Phi(0) = \mathbb{1} \end{cases}$$

- If $y(0) = y_0$, then $y(t) = \Phi(t)y_0$ and $y(T) = \Phi(T)y_0$



$\Downarrow$

Examine the eigenvalues of M to determine stability



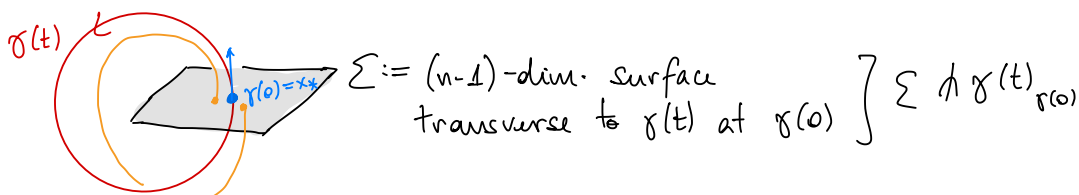
- if $\exists \mu$ with $|\mu| > 1 \Rightarrow$ instability
- if all but one μ have $|\mu| < 1 \Rightarrow$ stability

$$Mv = v \quad (v = \text{perturbation along } \sigma)$$

$$\Downarrow$$

$$\mu = 1$$

(*) POINCARÉ RETURN MAP



Define the map $P: \Sigma_0 \rightarrow \Sigma$

$$P(x_*) = x_*$$

$$P = \varphi_{\tau(x)}(x),$$

$$\tau(x) = 1^{\text{st}} \text{ return time to } \Sigma$$

LECTURE 9

Oct 28th 2025

STABLE & UNSTABLE MANIFOLDS

Recall:

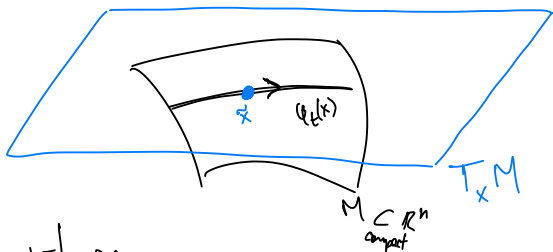
$$\frac{d\varphi_t(x)}{dt} = V(\varphi_t(x)) \quad , \quad \frac{dx(t)}{dt} = V(x(t)) .$$

REMARK: Linear non-autonomous equations that describe the stability (or lack thereof) of infinitesimal perturbations to the flow:

$$\frac{du(t)}{dt} = DV(\varphi_t(x)) u(t)$$

$u(t) \equiv$ infinitesimal perturbation to $\varphi_t(x)$ on some fixed orbit $\{\varphi_t(x)\}_{t \geq 0}$.

! HYPERBOLIC SYSTEMS: $\frac{d\varphi_t(x)}{dt} = V(\varphi_t(x))$ is a uniformly hyperbolic flow if $\exists C, \alpha > 0$ such that, at every x , there is a direct sum decomposition



where

$$\mathbb{R}^n = E^s(x) \oplus E^u(x) \oplus E^o(x)$$

not necessarily orthogonal

(STABLE SUBBUNDLE) $E^s(x) := \{u(0) \in T_x \mathbb{R}^n : \|u(t)\| \leq C e^{-\alpha t} \|u(0)\|, C, \alpha > 0\}$

(UNSTABLE SUBBUNDLE) $E^u(x) := \{u(0) \in T_x \mathbb{R}^n : \|u(t)\| \leq C e^{-\alpha |t|} \|u(0)\| \text{ as } t \rightarrow -\infty\}$

(CENTER SUBBUNDLE) $E^o(x) := \text{span } V(x)$

REMARK: $E^s \vee E^o \leftrightarrow$ perturbations that don't blow up in forward time

PROBLEM FOR MIDTERM: Show that changing $u(t)$ to $V(\varphi_t(x))$ does not show exponential growth or decay:

$$\frac{du(t)}{dt} = DV(\varphi_t(x)) u(t).$$

• Fixed point x_* such that $V(x_*) = 0 \Leftrightarrow \varphi_t(x_*) = x_* \quad \forall t$.
 Then x_* is hyperbolic whenever $\operatorname{Re}(\lambda) \neq 0 \quad \forall \lambda \in \operatorname{Spec}(DV(x_*))$.
 $\Leftrightarrow \exists \alpha > 0$ such that

$$\operatorname{Re}(\lambda) > \alpha \text{ or } \operatorname{Re}(\lambda) < -\alpha \quad \forall \lambda \in \operatorname{Spec}(DV(x_*))$$

NOTE: Perturbation system is now, for hyperbolic equilibria

$$\frac{du(t)}{dt} = DV(\varphi_t(x_*)) u(t) = \underbrace{DV(x_*)}_{\text{linear autonomous system}} u(t)$$

$$\Rightarrow u(t) = e^{tA} u(0)$$

$$\text{Thus: } \forall u(0) \in E^s, \quad \|u(t)\| \leq C e^{-\alpha t} \|u(0)\| \quad \text{as } t \rightarrow \infty$$

$$\forall u(0) \in E^u, \quad \|u(t)\| \leq C e^{-\alpha|t|} \|u(0)\| \quad \text{as } t \rightarrow -\infty$$

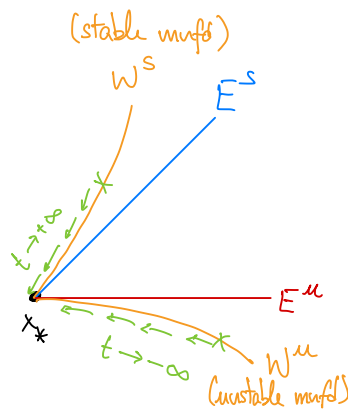
For fixed point: $E^0 = \{0\}$.

REMARK: $y(t) := \varphi_t(x) - x_*$. Then, as before,

$$\frac{dy}{dt} = Ay(t) + N(y(t))$$

$$V(x_*) = 0$$

$$A := DV(x_*)$$

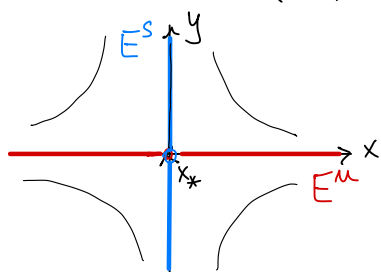


DEF: Global stable and unstable mfd's:

(GLOBAL STABLE MANIFOLD) $W^s(x_*) := \left\{ x \in \mathbb{R}^n : \|\varphi_t(x) - x_*\| \xrightarrow{t \rightarrow +\infty} 0 \right\}$

(GLOBAL UNSTABLE MANIFOLD) $W^u(x_*) := \left\{ x \in \mathbb{R}^n : \|\varphi_t(x) - x_*\| \xrightarrow{t \rightarrow -\infty} 0 \right\}$

EXAMPLE:



$$F(a,b) := \begin{bmatrix} z & 0 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \xrightarrow[\substack{\text{composition} \\ F^n \text{ n times}}]{\quad} \begin{bmatrix} z^n & 0 \\ 0 & \left(\frac{1}{3}\right)^n \end{bmatrix}$$

$x_* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is fixed pt.

$\rightarrow$ blows up $n \rightarrow \infty$
 $\downarrow$ 0 as $t \rightarrow \infty$

LECTURE 10

Oct 30th, 2025

- Hyperbolic Fixed Points: $\varphi_t(x_*) = x_* \Leftrightarrow V(x_*) = 0$

$$\operatorname{Re}(\lambda) \neq 0 \quad \forall \lambda \in \operatorname{Spec}(DV(x_*)),$$

- Uniformly Hyperbolic Fixed Point: presence of E^s, E^u, E^c exponential subspaces of $\mathbb{R}^n$ at every x with uniform (in x) rates of exponential expansion/contraction.

EXAMPLE: $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F(x_*) = x_*$. Suppose $\operatorname{Re}(\lambda) \neq 0$ for all $\lambda \in \operatorname{Spec} DF(x_*)$.

$$F \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2a \\ b/3 \end{pmatrix} \quad \hookrightarrow \quad F \doteq \begin{bmatrix} 2 & 0 \\ 0 & 1/3 \end{bmatrix}$$

$$F^n \doteq \begin{bmatrix} 2^n & 0 \\ 0 & (1/3)^n \end{bmatrix}. \quad \text{So, the flow here is}$$

$$V(x) = \begin{bmatrix} \overset{>0}{\log 2} & 0 \\ 0 & \log \frac{1}{3} \end{bmatrix} x \quad \left(\begin{array}{l} \text{LINEAR} \\ \text{HYPERBOLIC} \\ \text{DYNAMICS} \end{array} \right)$$

- $\log 2 > 0 \Rightarrow$ exponential growth in x_1
- $\log \frac{1}{3} < 0 \Rightarrow$ exponential decay in x_2



DEF: The unstable and stable manifolds for a fixed point
(can also be defined similarly for general points or even sets)

(STABLE MANIFOLD) $W^s(x_*) := \left\{ x \in \mathbb{R}^n : \left\| \varphi_t(x) - \underbrace{\varphi_t(x_*)}_{= x_* \text{ (fixed pt.)}} \right\| \xrightarrow{t \rightarrow +\infty} 0 \right\}$

(UNSTABLE MANIFOLD) $W^u(x_*) := \left\{ x \in \mathbb{R}^n : \left\| \varphi_t(x) - \underbrace{\varphi_t(x_*)}_{= x_* \text{ (fixed pt.)}} \right\| \xrightarrow{t \rightarrow -\infty} 0 \right\}$.

We can define the LOCAL stable and unstable manifolds for a hyperbolic fixed point as

(LOCAL STABLE MANIFOLD) $W_\delta^{loc, s}(x_*) := \left\{ x \in W^s(x_*) : \left\| \varphi_t(x) - x_* \right\| < \delta \quad \forall t \geq 0 \right\}$.

(LOCAL UNSTABLE MANIFOLD) $W_\delta^{loc, u}(x_*) := \left\{ x \in W^u(x_*) : \left\| \varphi_t(x) - x_* \right\| < \delta \quad \forall t \leq 0 \right\}$.

* REMARK: These manifolds are invariant

$$\varphi_t(W^s(x_*)) \subset W^s(x_*) \quad \forall t \geq 0$$

$$\varphi_t(W^u(x_*)) \subset W^u(x_*) \quad \forall t \leq 0$$

* LINEARIZATION: $\frac{du(t)}{dt} = DV(x_*)u(t) \rightsquigarrow u(t) = e^{tDV(x_*)}u(0)$

$$v \in E^s(x_*) \Rightarrow \|e^{tDV(x_*)}v\| \leq Ce^{-\alpha t} \|v\| \text{ as } t \rightarrow +\infty$$

$$v \in E^u(x_*) \Rightarrow \|e^{tDV(x_*)}v\| \leq Ce^{-\alpha|t|} \|v\| \text{ as } t \rightarrow -\infty.$$

Ex: (Same as before) $(0,0)$ is hyperbolic fixed pt.

$$F \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 1/3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \xrightarrow{\text{linear map}} DF(x_1, x_2) = \begin{pmatrix} 2 & 0 \\ 0 & 1/3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Exponential subspaces:

$$E^S(x_*) = x_2\text{-axis}$$

$$E^u(x_*) = x_1\text{-axis}$$

Manifolds:

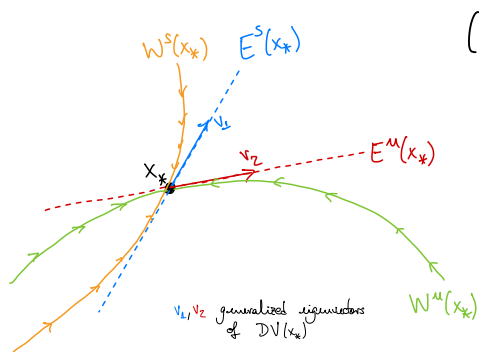
$$W^S(x_*) = x_2\text{-axis}$$

$$W^u(x_*) = x_1\text{-axis}$$



In this class, we only do this then for
hyperbolic fixed points

THM: (STABLE / UNSTABLE MANIFOLD THM) If V is a C^r vector field with a hyperbolic fixed point x_* ($V(x_*) = 0$), then



(i) $W^S(x_*)$ is a C^r manifold and its tangent space at x_* has the same dimension as the stable space $E^S(x_*)$ of x_*

(ii) $W^u(x_*)$ is a C^r manifold and its tangent space at x_* has the same dimension as the unstable space of x_*

(iii) $N^{\text{loc}, S}(x_*)$ is a Lipschitz graph over $E^S(x_*)$ (x, L(x))

LECTURE 11

Nov 4th 2023

OSELEDETS THEOREM AND LYAPUNOV EXPONENTS



STUDY THE PROOF OF STABLE / UNSTABLE MANIFOLD THEOREM
FOR MIDTERM

Highlights of proof: x_* hyperbolic fixed pt. $\Rightarrow \mathbb{R}^n = E^s(x_*) \oplus E^u(x_*)$

Set $y(t) := \varphi_t(x) - x_*$, $x \in B_\delta(x_*)$.

$$\begin{aligned} \frac{\partial y(t)}{\partial t} &= \frac{\partial \varphi_t(x)}{\partial t} = V(\varphi_t(x)) = V(y(t) + x_*) \\ &= \underbrace{DV(x_*) y(t)}_{\text{linear portion}} + \underbrace{N(y(t))}_{\text{nonlinear portion}} \end{aligned}$$

here, y is a function $[0, \infty) \rightarrow \mathbb{R}^n$. If $\|y(t)\|$ is bounded for all $t \geq 0$, then $y(t)$ satisfies

$$\frac{dy(t)}{dt} = DV(x_*) y(t) + N(y(t)) \quad (1)$$

provided that

$$\begin{aligned} y(t) &= e^{tA} (y(0))_s + \int_0^t e^{(t-z)A} (N(y(z)))_s dz \\ &\quad - \int_t^\infty e^{(t-z)A} (N(y(z)))_u dz \end{aligned} \quad (2)$$

$A := DV(x_*)$

For all $v \in \mathbb{R}^n$, since $\mathbb{R}^n = E^s(x_*) \oplus E^u(x_*)$, can write $v = v_s + v_u$, $v_s \in E^s(x_*)$ and $v_u \in E^u(x_*)$.

EXERCISE: Show that $y(t)$ satisfies (1).

Why is $y(t)$ bounded $\forall t \geq 0$?

Can $y(t) = \varphi_t(x) - x_*$ be unbounded? **Yes!**

If $y(t)$ is bounded, is the solution unique? **Yes!**

Claim: If $y(t)$ is bounded, the solution is unique.

Pf. (Contraction Mapping Theorem) Define

$$T: C^0([0, \infty), \mathbb{R}^n) \rightarrow C^0([0, \infty), \mathbb{R}^n)$$

$$T[y](t) := e^{tA}(y(0))_s + \int_0^t e^{(t-z)A} (N(y(z)))_s dz \\ - \int_t^\infty e^{(t-z)A} (N(y(z)))_u dz$$

Subclaim 1: T is well-defined (i.e., it maps C^0 fcts. to C^0).

Subclaim 2: Show that T is a contraction on the

subspace $V(\delta) := \{ t \mapsto y(t) : \|y\| \leq \delta \}$,

w.r.t. $\|y\| := \sup_{t \in [0, \infty)} |y(t)|$.

NTS: (i) $T: V(\delta) \rightarrow V(\delta)$

(ii) T contraction

Pf. (i) Note that

$$\|T[y](t)\| \leq \|e^{tA}(y(0))_s\| + \int_0^t \|e^{(t-z)A} N(y(z))_s\| dz$$

 If you're trying to prove uniqueness, there's usually a hidden contraction somewhere.

Hyperbolicity and definition
of stable/unstable vector

$$+ \int_t^\infty \|e^{(t-z)A} N(y(z))_u\| dz$$

$$\leq C e^{-\alpha t} \|y(0)\| + C \int_0^t e^{-\alpha(t-z)} \|N(y(z))_s\| dz \\ + C \int_\infty^t e^{-\alpha(t-z)} \|N(y(z))_u\| dz$$

Since $y \in V(\delta)$, we have $\|y\| \leq \delta \forall t$. Now, $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $\|N(v)\| < \varepsilon \|v\|$ whenever $\|v\| < \delta$. Thus,

$$\|T[y](t)\| \leq C e^{-\alpha t} \|y(0)_s\| + C\varepsilon \int_0^t e^{-\alpha(t-z)} \|y(z)\| dz \\ + C\varepsilon \int_\infty^t e^{-\alpha(t-z)} \|y(z)\| dz$$

$$\leq \underbrace{C e^{-\alpha t} \|y(0)_s\|}_{< \frac{\delta}{2}} + \underbrace{\frac{\tilde{C}\varepsilon\delta}{\alpha}}_{< \frac{\delta}{2}} \rightsquigarrow \varepsilon < \frac{\alpha}{2\tilde{C}}$$

So, take it to be

$$C e^{-\alpha t} \|y(0)_s\| < \delta/2.$$

Then $T: V(\delta) \rightarrow V(\delta)$, are desired.

□

Pf: (ii) T is a contraction b/c

$$\|T[y] - T[\tilde{y}]\| \leq e^{tA} \|y(0)_s - \tilde{y}(0)_s\| + (\dots) + (\dots)$$

$\leq$ Triangle Inequality

$$\dots < \frac{1}{2} \|y - \tilde{y}\| \text{ using same argument as above}$$

□

So, (i)+(ii) $\Rightarrow T$ is a contraction on a complete space

$\Rightarrow T$ has a unique fixed point in $V(\delta)$.

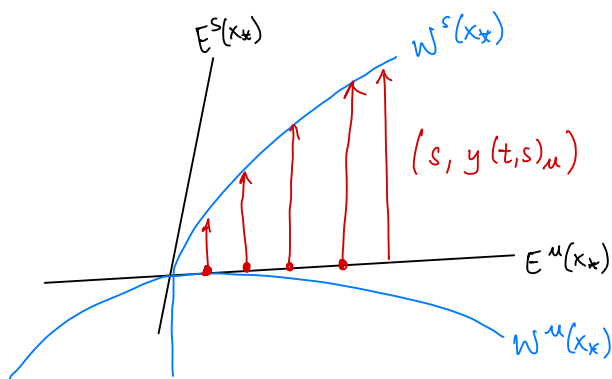
Upshot: If y is bounded $\forall t \geq 0$ and solves (1), then y is unique.

lastly: Use Generalized Grönwall to get that

$$\|y(t)\| \leq \tilde{C} e^{-\tilde{\alpha}t} \|y(0)_s\|.$$

Remark: Setting $s := y(0)_s$

$$\|y(t, s) - y(t, \tilde{s})\| \leq K \|s - \tilde{s}\| \Rightarrow y \text{ is a Lipschitz graph over } E^s(x_*).$$



INTUITION: Stable / Unstable Mnf! Then gives us a new coordinate system to write vectors in

$$n_s := \dim E^s(x_*) = \dim W_{loc}^s(x_*)$$

$$n_u := \dim E^u(x_*) = \dim W_{loc}^u(x_*)$$

$$n = n_s + n_u$$

$$\mathbb{R}^n = E^s(x_*) \oplus E^u(x_*).$$

Namely: $(s, y(t, s))$

n_s entries

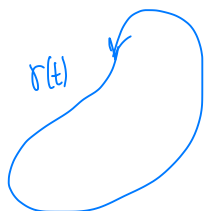
n_u entries

$$n_s + n_u = n$$

LECTURE 12

Nov 9, 2025

STABILITY PROPERTIES OF CYCLES:



$$\dot{x} = V(x, c), \quad x \in \mathbb{R}^n$$

↖ parameter $c \in \mathbb{R}$

$$\dot{\gamma}(t) = V(\gamma(t))$$

$$\gamma(t) = \gamma(t+T)$$

LINEAR STABILITY OF $\gamma(t)$: $x(t) = \gamma(t) + y(t)$

$$\Rightarrow \dot{\gamma} + \dot{y} = V(\gamma + y) = V(\gamma(t)) + \underbrace{DV(\gamma(t))}_{A(t)} y + \text{h.o.t.}$$

↖ Jacobian

Linearized Problem: $\dot{y} = \underbrace{A(t)}_{\uparrow} y$

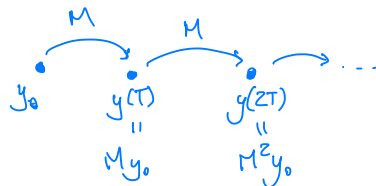
NOTE: $\gamma(t+T) = \gamma(t) \Rightarrow A(t+T) = A(t)$.

Fundamental Matrix Solution: $\dot{\Phi}(t) = A(t) \Phi(t)$, $\Phi_0 = \mathbb{I}$.

↖ "Propagator"

If $y(0) = y_0$, then $y(t) = \Phi(t) y_0$

Monodromy Matrix: $\Phi(T) := M$



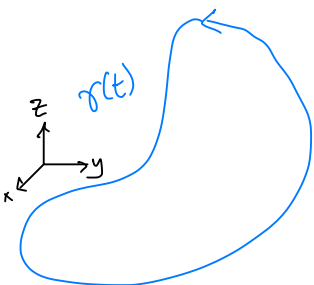
→ Eigenvalues of M :

$$\mu_1 = 1, \mu_2, \dots, \mu_n.$$

← If $|\mu_i| > 1$ for some $i \geq 2$, then perturbations actually GROW

(HW03)

EXAMPLE: (ROESSLER ATTRACTOR) $a = b = 0.2$ and $c = 2.5$.



$$\begin{cases} \dot{x} = -y - z \\ \dot{y} = x + ay \\ \dot{z} = b + z(x-c) \end{cases}$$

$$DV(x, y, z) = \begin{bmatrix} 0 & -1 & -1 \\ 1 & a & 0 \\ z & 0 & x-c \end{bmatrix}$$

Consider the 6-dim system

$$\begin{cases} \dot{x} = -y - z \\ \dot{y} = x + ay \\ \dot{z} = b + z(x-c) \\ \delta \dot{x} = -\delta y - \delta z \\ \delta \dot{y} = \delta x + a \delta y \\ \delta \dot{z} = z \delta x + \delta z(x-c) \end{cases}$$

WANT: Monodromy Matrix
 $\Phi(T) =: M$

Initial conditions

$$(x_0, y_0, z_0, 1, 0, 0)$$

$$(x_0, y_0, z_0, 0, 1, 0)$$

$$(x_0, y_0, z_0, 0, 0, 1)$$

GOAL: Choose x_0, y_0, z_0 so that they are on the limit cycle!

Flows thru by T : Need to choose x_0, y_0, z_0 on the cycle?

$$\varphi_T(x_0, y_0, z_0, 1, 0, 0) = (x_0, y_0, z_0, a, b, c)$$

$$\varphi_T(x_0, y_0, z_0, 0, 1, 0) = (x_0, y_0, z_0, d, e, f)$$

$$\varphi_T(x_0, y_0, z_0, 0, 0, 1) = (x_0, y_0, z_0, g, h, k)$$

$$\Rightarrow M = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix}$$

Verify this is a good M :

(i) Check that one of the eigenvalues is ≈ 1 .

(ii) Check that the eigenvector is tangent to the limit cycle.

(iii) ABEL'S IDENTITY:

$$\det(\Phi(t)) = \exp \left[\int_0^t \text{tr}(A(s)) ds \right]$$

$$\leadsto \det(M) = \exp \left[\int_0^T (\alpha + x(s) - c) ds \right] \stackrel{!}{=} \underset{\uparrow}{\mu_2 \mu_3} \cdot \underset{\mu_1}{\frac{1}{\mu_1}} \stackrel{!}{>} 0$$

Verify that this is the case

FIND T: Times the time series crosses zero



Make sure we HAVE A SINGLE PEAK (aka 1-cycle)

CHECK BY COUNTING PEAKS IN TIME SERIES

* APPROXIMATING STABLE / UNSTABLE MANIFOLDS

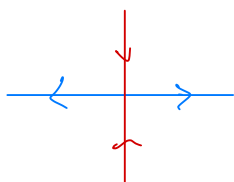
Ex:
$$\begin{cases} \dot{x} = 2x + y^2 \\ \dot{y} = -2y + x + y^2 \end{cases}$$

linearize about origin $\rightsquigarrow$

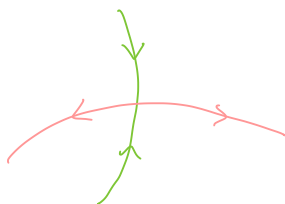
$DV(0)(x,y) = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$

$\begin{matrix} \vdots \\ 4 \end{matrix}$

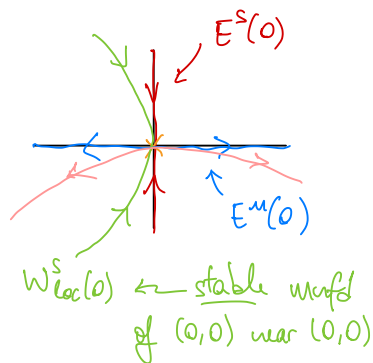
(HW03 $\rightarrow$ find $W_{loc}^u(0)$)



(linearization)



(nonlinear)



Use Stable/Unstable Mnfld Thm to find $W_{loc}^s(0)$:

- Recall $W_{loc}^s(0)$ is a Lipschitz graph over $E^s(0)$; i.e.,

$$W^s(0) = \{ (x,y) = (\underline{h_s(y)}, y) : |y| \leq \delta \}$$

(1) $h_s(0) = 0 \Leftarrow W_{loc}^s$ is a graph over E^s

(2) $h'_s(0) = 0 \Leftarrow$ and it's tangent to E^s at $(0,0)$

(3) If $(x_0, y_0) = (h_s(y_0), y_0)$ is the initial condition, then

$$(x(t), y(t)) = (h_s(y(t)), y(t))$$

So $x(t) = h_s(y(t)) \rightsquigarrow \dot{x} = h'_s(y(t)) \dot{y}$

(Using the equations)

$$\rightsquigarrow (2x + y^2)|_{x=h_s} = h'_s(y(t)) (-2y + x + y^2)|_{x=h_s}$$

$x = h_s$

$$\leadsto \begin{cases} (2h_s + y^2) = h'_s (-2y + h_s + y^2) \\ h'_s(0) = h_s(0) \end{cases}$$

valid for $|y|$
suff. small

⚠ GOAL: Approximate h_s using a power series

$$h_s(y) = \alpha_1 + \alpha_2 y + \alpha_3 y^2 + \alpha_4 y^3 + \dots$$

Substitute this into the eq.:

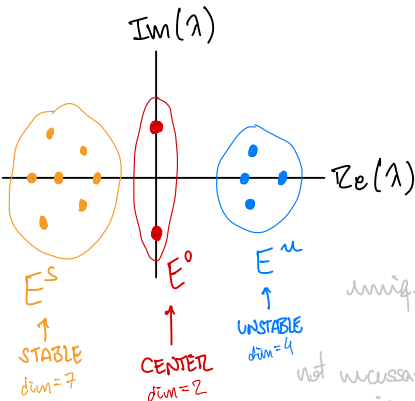
$$2\alpha_2 y^2 + 2\alpha_3 y^3 + \dots + y^2 = (2\alpha_2 y + 3\alpha_3 y^2 + \dots) \cdot (-2y + \alpha_2 y^2 + \alpha_3 y^3 + \dots + y^2)$$

Take terms up to $O(y^2)$: $2\alpha_2 + 1 = -4\alpha_2 \Rightarrow \alpha_2 = -\frac{1}{6}$.

NON - HYPERBOLIC CASE: CENTER MANIFOLD

$$\begin{cases} \dot{x} = V(x), & V \in C^k, k \geq 1 \\ V(0) = 0 & \leftarrow \text{wlog} \end{cases}$$

Let $\lambda :=$ eigenvalues of $DV(0)$



• There is a neighborhood of the origin where $\exists$ locally C^k invariant manifolds:

$$\begin{cases} W_{loc}^s = \text{tangent to } E^s, |x(t)| \rightarrow 0 \text{ as } t \rightarrow +\infty \\ W_{loc}^u = \text{tangent to } E^u, |x(t)| \rightarrow 0 \text{ as } t \rightarrow -\infty \\ W_{loc}^0 = \text{tangent to } E^0 \end{cases}$$

unique (for the first two)
not necessarily unique (for the third)

LECTURE 13

CHAOTIC SYSTEMS

Nov 11, 2025

OSELEDETS MULTIPLICATIVE ERGODIC THEOREM (MET)

* LINEARIZATION EXPLAINING BEHAVIOR OF FULL NONLINEAR SYSTEM

(Discrete Time)

MAP: $F: M \subset \mathbb{R}^n \rightarrow M \subset \mathbb{R}^n$, M compact.

LINEARIZED DYNAMICS: $DF(x): \mathbb{R}^n \rightarrow \mathbb{R}^n$. For an orbit $\{x_0, x_1, \dots\}$

s.t. $x_i = F(x_{i-1})$. Then

$$\text{Linear in } v, \text{ nonautonomous} \rightarrow v_t = DF(x_{t-1}) v_{t-1}, \quad \begin{cases} v_0 = \text{infinitesimal perturbation at } x_0 \\ v_t = \text{infinitesimal perturbation at } x_t. \end{cases}$$

- if x is a fixed point $\Rightarrow$ can use Floquet theory as before to study the dynamics.

- if x is not a fixed point: $v_t \in \mathbb{R}^n$, $v_t = A_{t-1} v_{t-1}$

$$\Rightarrow v_t = \underbrace{A_{t-1} A_{t-2} \dots A_1 A_0}_{B(t)} v_0$$

$\uparrow A_t := DF(x_t)$

ASSUME: $A_t \in \mathbb{R}^{n \times n}$ $\forall t$ is nonsingular and $\frac{1}{t} \log \|A_t\| \xrightarrow{t \rightarrow \infty} 0$.

should be subexponential

LYAPUNOV EXPONENT: $\mathbb{R} \ni \lambda(x_0, v) := \lim_{t \rightarrow \infty} \frac{1}{t} \log \|B(t)v\|$.

existence/uniqueness $\Rightarrow$ fixing IC x_0 fixes the orbit
and $B(t) = A_{t-1} \cdots A_0$, $A_0 = DF(x_0)$

$\rightarrow$ The rate of the exponential growth of $\|B(t)v\|$ as $t \rightarrow \infty$
 $\Leftrightarrow \lambda(x_0, v)$ is a scalar that gives the rate of asymptotic exponential growth / decay of infinitesimal linear perturbations along a reference orbit of x_0 in the direction of v at x_0

$$\|v_t\| = \|B_t(x_0)\| \approx O(e^{\lambda(x_0, v)t} v) \text{ as } t \rightarrow \infty$$

OSSELEDETS THM: (1) The Lyapunov exponents $\lambda(x_0, v)$ only take a finite number $p \in \mathbb{N}$ of distinct values $p \leq n$ for any v .

Let $\{\lambda_i\}_{i=1}^p$, $p \leq n$, be the Lyapunov exponents ordered as

$$\lambda_1 > \lambda_2 > \cdots > \lambda_p$$

and multiplicity k_i of λ_i , $i = 1, \dots, p$. Then,

$$\sum_{i=1}^p k_i \lambda_i = \lim_{t \rightarrow \infty} \frac{1}{t} \log |\det B(t)|.$$

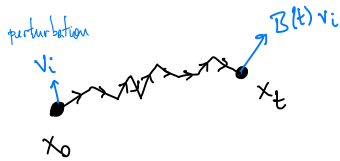
$$(2) \lim_{t \rightarrow \infty} \left[(B(t))^T B(t) \right]^{1/2t} = \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_p})_{n \times n}$$

$\uparrow$
multiplicity...

NOTE: $B(t) = U(t) \Sigma(t) V(t)^T$ (SVD)

$$\Rightarrow B(t) v_i = \sigma_i u_i$$

$\nwarrow$ right sing. vec.
 $\swarrow$ left sing. vec.



$$\Rightarrow \text{vol}_t(\kappa) = \text{vol}_0(\kappa) \cdot \left(\sum_{i=1}^{\kappa} \sigma_i \right)^t \quad (\text{discrete time})$$

$$\text{vol}_t(\kappa) = \text{vol}_0(\kappa) \cdot \exp \left[\sum_{i=1}^{\kappa} \sigma_i \right] \quad (\text{continuous time})$$

EXAMPLE: $V_0 := \{0\}$, then $B(t)V_0 = \{0\}$.

$$\text{So } \lambda(v) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \|B(t)v\| = \lim_{t \rightarrow \infty} \frac{1}{t} \log(0) = -\infty.$$

$V_1 := \{v \in \mathbb{R}^n : \lambda(v) \leq \lambda_1\} \cong \mathbb{R}^n$ b/c λ_1 is the largest Lyapunov exponent

$V_2 := \{v \in \mathbb{R}^n : \lambda(v) \leq \lambda_2\}$ is $(n - n_1)$ -dimensional

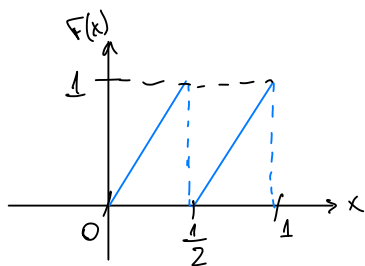
$$\mathbb{R}^n \cong V_1 \supset V_2 \supset V_3 \supset \dots \supset V_p \supset V_{p+1} \dots$$

$\vdots \{0\}$

DEF: In a chaotic dynamical system, there is at least one positive Lyapunov exponent $\lambda_1 > 0$. $\Rightarrow V_i \neq \emptyset$

EXAMPLE: (1-dim chaos) For $x \in [0, 1]$,

$$F(x) := \begin{cases} 2x, & x < 1/2 \\ 2x-1, & x > 1/2 \end{cases}, \quad x_t = F(x_{t-1})$$



$x_* = 0$ is a fixed point



$$A := DF(x_*) = 2$$

Then, infinitesimal perturbation system $v_{t+1} = DF(x_*) v_t$

$$v_{t+1} = 2v_t, \quad v_0 \neq 0, \|v_0\| = \varepsilon$$

Then, the Lyapunov exponent of v is

$$\lambda(v) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \|B(t)v\| = \log 2 > 0 \Rightarrow \text{CHAOS}$$

Orbits: $x = 0 \rightarrow$ fixed pt.

$x = 0.1 \rightarrow 0.1, 0.2, 0.4, 0.8, 0.6, 0.2, 0.4, \dots$ (periodic)

$x = 0.1 + \varepsilon \rightarrow 0.1 + \varepsilon, 0.2 + \varepsilon', 0.4 + \varepsilon'', \dots$ (discrepancy although they exist on $[0, 1]$)

LECTURE 14

Nov 13th, 2025

CHAOS $\leftrightarrow$ sensitivity to ICs & loss of memory of IC.

Ex: Smale Horseshoe } Chaos coming from deterministic systems
Baker unfolded map

EXAMPLE: (Baker Unfolded Map)

$$F(x,y) = \begin{pmatrix} 2x - L2x \\ y/2 - L2x/2 \end{pmatrix}$$



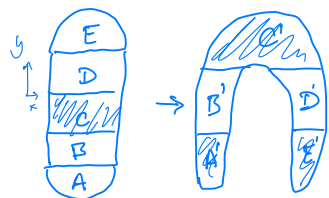
SYMBOLIC DYNAMICS: $x \in [0,1]$ in binary: $0.01\dots$

times 2 $\Leftrightarrow 0.1\dots$ (left shift)

divide by 2 $\Leftrightarrow 0.001\dots$ (right shift)

NOTE: Smale Horseshoe dynamics on

$$\Delta := \left\{ (x,y) \in B \cup D : F_{SH}^n(x,y) \in B \cup D \right\}$$



Attractor of the Smale Horseshoe
 F_{SH} map is $\simeq C \times C$, $C \equiv$ middle
third
Cantor
set

Removing middle
thirds

OSELEDETS THEOREM: Deals w/ behavior of linearizations
 Fix orbit $\{\varphi_t(x)\}_{t \geq 0}$, and

$$\frac{d\varphi_t(x)}{dt} = V(\varphi_t(x)) \rightsquigarrow \frac{du}{dt} = DV(\varphi_t(x)) u(t)$$

Suppose we have a discrete time map F for the dynamics.

$$M(0) \in \mathbb{R}^{n \times n}, \quad \frac{dM(t)}{dt} = DV(\varphi_t(x)) M(t)$$

$$M(t) = \underbrace{F(t)} M(0)$$

tangent propagator $\leftrightarrow$ fundamental matrix solution

$$u(t) = \underbrace{F(t)}_{\substack{\in \\ \mathbb{R}^{n \times n}}} u(0) \quad \text{Take SVD of } F(t):$$

$$F(t) = U(t) \Sigma(t) V(t)^T$$

$$F(t) v_i(t) = \sigma_i(t) u_i(t)$$

$$O(t) := \lim_{t \rightarrow \infty} \left(F(t)^T F(t) \right)^{1/2t} = \lim_{t \rightarrow \infty} \left(V(t) \Sigma(t)^2 V(t)^T \right)^{1/2t}$$

$$\left[V(t) \xrightarrow{t \rightarrow \infty} \text{matrix of eigenvectors of } O(t) \right]$$

$$= \lim_{t \rightarrow \infty} V(t) \Sigma(t)^{1/t} V(t)^T$$

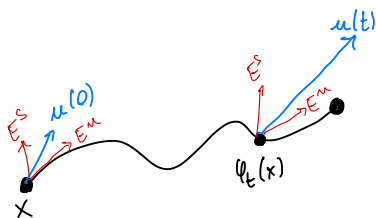
$$= \text{diag} \left(e^{\lambda_1}, \dots, e^{\lambda_p} \right)_{n \times n} \leftarrow \begin{array}{l} \text{multiplicity} \\ \text{of Lyapunov} \\ \text{exponents} \end{array}$$

$\uparrow \quad \uparrow \quad \uparrow$
 exp. of Lyapunov exponent

$\lambda_1 > \dots > \lambda_p$ w/ multiplicity

NOTE: $\lambda_i(t) = \frac{1}{t} \log \sigma_i(t)$

OMET: as $t \rightarrow \infty$, $\frac{1}{t} \log(\sigma_i(t))$ converges as $t \rightarrow \infty$



$u(0) \sim$ randomly chosen on $\mathbb{R}^n$

$$\|u(t)\| \sim O(e^{\lambda_1 t})$$

i.e., $\lambda_1 > 0 \Rightarrow \varphi_t$ IS CHAOTIC

$$\frac{du(t)}{dt} = DV(\varphi_t(x)) u(t)$$

COMPUTE λ_1 : $\frac{du(t)}{dt} = A(t) u(t)$. Then λ_1 is approx. the rate of growth $\|u(t)\| \sim O(e^{\lambda_1 t})$ for each initial condition. That is,

$$\lambda_1 \approx \frac{1}{t} \log \frac{\|u(t)\|}{\|u(0)\|}$$

NUMERICAL ERROR: be careful with taking logs/dividing by, e.g., e^{100} b/c that's basically 0.

HW04: $\mathbb{R}^6 = E^u \oplus E^s \oplus E^o$

positive Lyapunov exponents
" $\dim E^u$

There's always a zero Lyapunov exponent b/c v itself satisfies the linearized equation $\Rightarrow v \in E^o$

$\frac{d\Phi}{dt} = A(t)\Phi, \Phi(0) = I_d, A(t) = DV(x_t)A$
OMET: let $V \in C^\infty$ and say $\Phi(t)$ is volume preserving.
 Then, the limits $\text{sing. vals. of } \Phi(t)$

i.e., $\det \Phi(t) = 1 \forall t$

$\Leftrightarrow \text{div } V = 0$

(e.g. Hamiltonian systems by Liouville's Thm)

$\lim_{t \rightarrow \infty} \frac{1}{t} \log \sigma_i(t) = \lambda_i$ exist

and the orbits behave as $\|u(t)\| \approx O(e^{\lambda_1 t})$,

$\lambda_1 = \max LE = \lim_{t \rightarrow \infty} \frac{1}{t} \log \sigma_1$

————— //

FACTS: 1) $\lambda_1 \approx \frac{1}{t} \log \left(\frac{\|u(t)\|}{\|u(0)\|} \right)$

2) LES:

• Continuous systems: $\lambda(x_0, v) := \lim_{t \rightarrow \infty} \frac{1}{t} \log \|\Phi(t)v\|$

• Discrete systems: $x_{n+1} = f(x_n)$

$\lambda(x_0) := \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \log |f'(x_i)|$

LECTURE 15

BIFURCATION THEORY

Nov 20, 2025

POINCARÉ 1885: Topological change in the structure of a dynamical system when a parameter crosses a bifurcation value.

GOAL: LOCAL BIFURCATIONS $\rightarrow \dot{x} = V(x; \lambda)$ $\leftarrow C^r, r \geq 1$
local to a $\overset{\text{non-hyperbolic}}{V}$ equilibrium x_0 and occurring at a value λ_0
 $\leftarrow$ parameters, $x \in \mathbb{R}^n, \lambda \in \mathbb{R}^k$

REMARK: If x_0 is hyperbolic, then small changes of λ do not change $\dim E^s(x_0)$ or $\dim E^u(x_0)$

$\Rightarrow$ stability doesn't change (equilibrium x_0 cannot be destroyed).

$\Rightarrow$ Hyperbolic equilibria are ROBUST. $\leftarrow \Leftrightarrow$ structurally stable

$\Rightarrow$ Need to study non-hyperbolic equilibria for bifurcation

DEF: STEADY STATE BIFURCATION $\Leftrightarrow$ creation or destruction of equilibria.

IMPLICIT FUNCTION THEOREM: If $DV(x_0, \lambda_0)$ is invertible (i.e., $\det DV(x_0, \lambda_0) \neq 0$), then $\exists V \ni x_0, W \ni \lambda_0$, and a unique C^r function $\xi(\lambda): W \rightarrow V$ for which

$$V(\xi(\lambda), \lambda) = 0, \text{ and } x_0 = \xi(\lambda_0)$$

 If $\det D_x V(x_0, \lambda_0) \neq 0$, then we can use linear approximation to approximate $\xi(\lambda)$:

$$0 = V(x, \lambda) = V(x_0, \lambda_0) + D_x V(x_0, \lambda_0)(x - x_0) + D_\lambda V(x_0, \lambda_0)(\lambda - \lambda_0) + \dots$$

$$\Rightarrow x \approx x_0 - [D_x V(x_0, \lambda_0)]^{-1} D_\lambda V(x_0, \lambda_0)(\lambda - \lambda_0)$$

NON-HYPERBOLIC MARTMAN-GROBMAN: Let

$$(*) \begin{cases} \dot{x} = Cx + F(x, y, z) & \text{(center directions)} \\ \dot{y} = Sy + G(x, y, z) & \text{(stable directions)} \\ \dot{z} = Vz + H(x, y, z) & \text{(unstable directions)} \end{cases}$$

F, G, H are $\mathcal{O}(x, y, z)$. Then, $\exists N \ni 0$ s.t. ← wlog via translations

$$W_{loc}^c(0,0,0) = \{(x, g(x), h(x)) : x \in E^c\} \cap N,$$

and the dynamics of $(*)$ are topologically conjugate to

$$\dot{x} = Cx + F(x, g(x), h(x)), \quad \dot{y} = Sy, \quad \dot{z} = Vz$$



Dimension reduction: can just ignore the "nice" directions that are hyperbolic (E^u, E^s) and just focus on the problematic non-hyperbolic directions (E^c).

BIFURCATION PROBLEM: Consider the extended 2-dim. center mfd (2-dim b/c $x \in \mathbb{R}, \lambda \in \mathbb{R}$).

$$\begin{cases} \dot{x} = 0 \cdot x + \underline{F(x, \lambda, g(x, \lambda), h(x, \lambda))} =: f(x, \lambda) \\ \dot{\lambda} = 0 \cdot \lambda \leftarrow \lambda \text{ is a parameter (const.)} \end{cases}$$

[; don't need the $\dot{y}, \dot{z}$
b/c of non-hyperbolic HGF]

Thus: $\dot{x} = f(x, \lambda), \quad x \in \mathbb{R}, \lambda \in \mathbb{R}$.

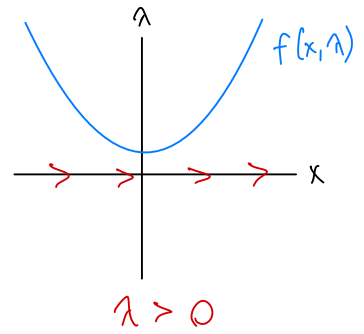
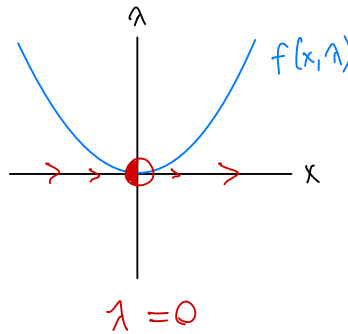
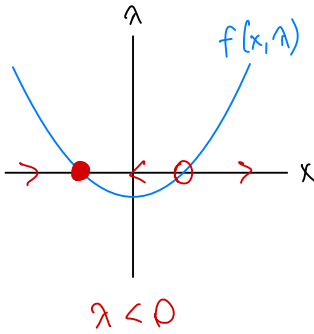
Assume equilibrium at $x=0$ and bifurcation at $\lambda=0$.

$$\left. \begin{aligned} f(0,0) &= 0 \\ \frac{\partial f}{\partial x}(0,0) &= \mu = 0 \end{aligned} \right\} \begin{array}{l} \text{DEFINING PROPERTIES} \\ \text{OF STEADY-STATE BIFURCATION} \end{array}$$

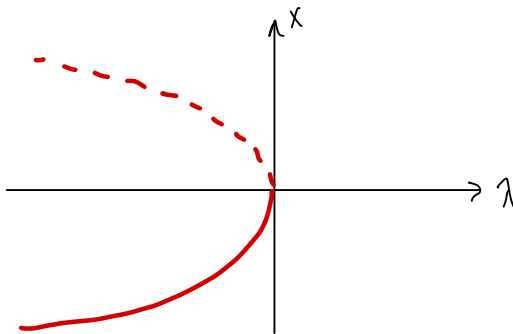
Taylor Expand in (x, λ) :

$$\begin{aligned} f(x, \lambda) &= \cancel{f(0,0)}^{=0} + \cancel{\frac{\partial f}{\partial x}(0,0)}^{=0} x + \overbrace{\frac{\partial f}{\partial \lambda}(0,0)}^{\text{assume } \neq 0} \lambda + \frac{1}{2} \overbrace{\frac{\partial^2 f}{\partial x^2}(0,0)}^{\text{assume } \neq 0} x^2 \\ &\quad + \frac{\partial^2 f}{\partial x \partial \lambda}(0,0) x \lambda + \frac{1}{2} \frac{\partial^2 f}{\partial \lambda^2}(0,0) \lambda^2 \end{aligned}$$

SADDLE - NODE BIFURCATION: Equilibria appear/disappear in pairs



SADDLE - NODE BIFURCATION



——— STABLE
- - - - - UNSTABLE

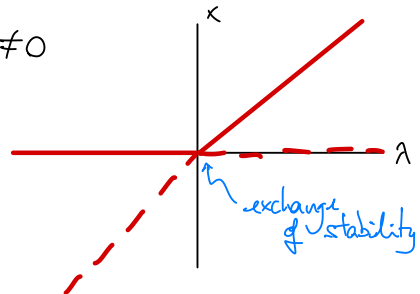
——— " ———

TRANSCRITICAL BIFURCATION: Always 2 equilibria that change stability at the bifurcation value (one of the eq. is always at 0)

$$\frac{\partial f}{\partial \lambda}(0,0) = 0, \quad \frac{\partial^2 f}{\partial x^2}(0,0) \neq 0, \quad \frac{\partial^2 f}{\partial x \partial \lambda}(0,0) \neq 0$$

$$\dot{x} = f(x, \lambda) = x g(x, \lambda)$$

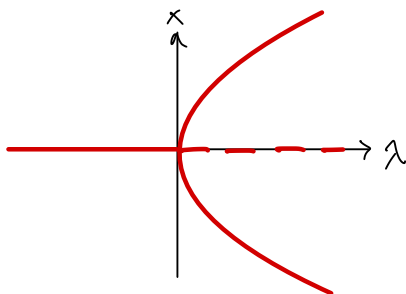
Ex: Logistic $\dot{x} = x(\lambda - x)$



PITCHFORK BIFURCATION: When we have $x \mapsto -x$ symmetry;
 i.e., $x = \tilde{x}(t)$ solution to $\dot{x} = f(x) \Rightarrow x = -\tilde{x}(t)$ also solution
 $\Rightarrow f(\tilde{x}) = -f(-\tilde{x})$ (f odd in x)

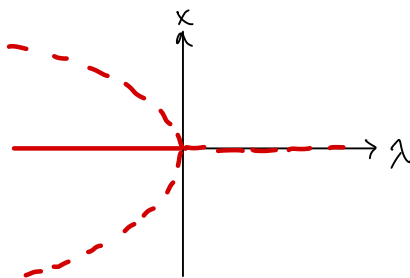
1) SUPERCritical PITCHFORK

$$\dot{x} = \lambda x - x^3$$

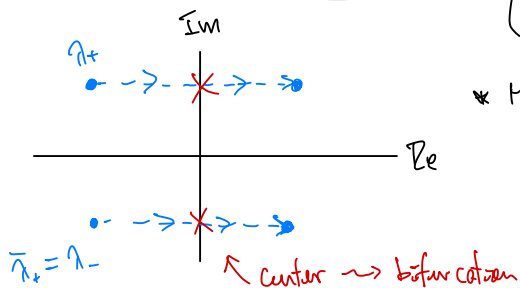


2) SUBCRITICAL PITCHFORK

$$\dot{x} = \lambda x + x^3$$



HOPF BIFURCATION:



$$\text{(Normal Form)} \begin{cases} \dot{x} = -\omega y + p(x, y) \\ \dot{y} = \omega x + q(x, y) \end{cases}$$

* Happens at $\mu_x = 0$ if

1) $\omega \neq 0$

2) $\frac{d}{d\mu} \text{Re}(\lambda_{\pm}) \neq 0$ ("horizontal" speed of the evs)

→ Supercritical : $\alpha < 0$ (stable cycle created as $\mathbb{R}(\lambda_1) \rightarrow \text{right}$)

→ Subcritical : $\alpha > 0$ (unstable cycle created as $\mathbb{R}(\lambda_1) \rightarrow \text{left}$)

In $\mathbb{R}^d$:

$$\alpha = \frac{1}{16} [P_{xxx} + P_{xyy} + q_{xxy} + q_{yyx}]$$

$$- \frac{1}{16\omega} [q_{xy}(q_{xx} + q_{yy}) - P_{xy}(P_{xx} + P_{yy}) + P_{xx}q_{xx} - P_{yy}q_{yy}]$$

————— // —————

HOMOLOGICAL OPERATOR: $L_A: H^n_k \rightarrow H^n_k$ linear

↑ homogeneous polys. in $\dim = n$ of $\deg = k$

$$L_A(h) := (Dh)A - Ah$$

Basis for H^n_k is $\{x^\alpha\}$, $\alpha \in \mathbb{N}^n$ multi-index with $|\alpha| = k$.